Mentify

AI Enhanced Digital Learning Platform

Software Requirement Specification (SRS)

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01	2026/03/14	Project Team	Initial draft created
02	2026/10/24	Project Team	Revised to align with the implemented Mentify system, including updates to functional and non-functional requirements, system scope, architecture, workflows, diagrams, and supporting documentation. Deferred requirements were identified and documented.

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Chapter 1: Introduction

The Mentify Digital Learning Platform is a comprehensive Learning Management System (LMS) designed and developed for TIT Education, an educational institution that provides tailored tuition services for students. This document presents the complete Software Requirement Specification (SRS) for the system, capturing all relevant functional and non-functional requirements, use cases, activity flows, system constraints, and feasibility considerations. In this revision, the document has been aligned with the system as actually developed, so that every requirement either describes implemented behaviour or is clearly identified as deferred or not yet verified.

Furthermore, this document has been prepared in accordance with the guidelines of the SENG 31242 module and the SENG 34213 System Development Project guidelines. Where the design evolved during development, the changes are recorded in the Requirement Change Register (Appendix A) and the Requirements Traceability Matrix (Appendix B), and are to be supported by Architectural Decision Records (ADRs) in the Software Design Specification (SDS).

1.1 Purpose

The purpose of this document is to provide a detailed and structured specification of the requirements for the Mentify Digital Learning Platform. It is intended to serve as the primary reference for the development team, the client (TIT Education), project supervisors, and all stakeholders involved in the design, development, testing, and deployment of the system. This document defines the system's functional capabilities, non-functional quality attributes, user interactions, business rules, and technical constraints.

Mentify is specified as a multi-role, microservice-based educational platform. Its scope covers academic administration, student and teacher management, grade-based course organisation, course enrolment, learning resources, assignments, quizzes, examinations, attendance, scheduling, communication, course payments, role-authorised AI assistance, and reporting.

The intended audience of this document includes software engineers, UI/UX design-

ers, quality assurance engineers, the project supervisor, and the client representative at TIT Education.

1.2 Scope

System Name: Mentify – AI Enhanced Digital Learning Platform

Mentify is a web-based Learning Management System (LMS) developed to support the academic and administrative activities of TIT Education. The system provides role-based functionality for Students, Teachers, Administrators, and Super Administrators. Its core capabilities include user account management, authentication and authorisation, grade and course management, student enrolment, educational content management, assignments, quizzes, examinations, attendance, communication, calendar management, academic analytics, notifications, and financial transactions associated with course enrolment.

Students can register through administrator-provisioned accounts, browse courses, pay applicable course fees, request enrolment, access authorised learning materials, submit assignments, attempt multiple-choice quizzes, view examination information and results, track attendance, and communicate through course groups. Teachers manage the courses assigned to them, upload materials, design assignments and quizzes, grade submissions with written feedback, record attendance, manage examination marks, and monitor student performance through analytics. Administrators manage user accounts, grades and grade-based monthly fees, courses, enrolments, payments and refunds, materials, marketplace products, and institutional analytics. Super Administrators hold the highest administrative authority, including the creation of administrator accounts.

An integrated AI service supports educational content summarisation, AI-assisted quiz draft generation, and role-authorised conversational analysis of academic and administrative information. Conversational AI is available to Teachers, Administrators, and Super Administrators, while lesson and material summarisation is accessible to authorised learners and staff.

The platform integrates a payment service based on Stripe for processing eligible course payments. Students may initiate payments and view their own payment information, while authorised administrative users may review transactions and manage refunds. Mentify additionally provides an administrator-managed digital educational material marketplace catalogue through which published learning resources can be discovered.

The backend follows a microservice architecture using Java and Spring Boot, PostgreSQL, Spring Cloud components (API Gateway, Config Server, Eureka), Kafka-based event communication, Redis-enabled shared infrastructure, and S3-compatible object storage.

Explicit Exclusions and Deferred Functionality:

The system shall not include live video conferencing or real-time virtual classroom functionality in the current release. Offline access to course content is not supported. Integration with external Student Information Systems (SIS) of other institutions is not considered within the scope of this implementation. Payroll, institutional accounting, and recurring subscription billing are outside the demonstrated scope; the grade-based monthly fee is a configured fee value and does not by itself constitute an automated recurring billing engine.

The following capabilities that appeared in the original draft of this document are not established in the implemented system and are therefore classified as deferred: a dedicated Support Team role and helpdesk ticket portal; a student-facing AI help chatbot with ticket escalation; AI-generated personalised course recommendations; AI-assisted scoring of assignments; third-party (Google) social authentication, Single Sign-On, and Multi-Factor Authentication; custom user-defined roles; short-answer quiz questions; and an end-to-end paid-material checkout with purchase entitlement. Each is recorded in Appendix A so that original commitments remain traceable.

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
LMS	Learning Management System, a software platform used for delivering, managing, and tracking educational content.
AI	Artificial Intelligence, a technology that enables automated decision-making, content generation, and personalisation.
JWT	JSON Web Token, a compact and URL-safe token used for secure authentication between client and server.
JWKS	JSON Web Key Set, a published set of public keys used to verify the signature of issued JWTs.
OTP	One-Time Password, a short-lived code sent to a user to verify identity during account recovery.

Term	Definition
SSO	Single Sign-On, an authentication mechanism that allows users to log in once and access multiple systems.
MFA	Multi-Factor Authentication, a security method that requires two or more verification factors during login.
RBAC	Role-Based Access Control, a method of restricting system access based on user roles.
MCQ	Multiple Choice Question, a question format consisting of predefined answer options.
SRS	Software Requirement Specification, a document that describes the functional and non-functional requirements of a system.
SDS	Software Design Specification, a document that describes the design of a system that satisfies the SRS.
ADR	Architectural Decision Record, a short document recording a significant architectural decision, its context, alternatives, and consequences.
NFR	Non-Functional Requirement, a requirement that specifies system quality attributes such as performance, security, and usability.
FAQ	Frequently Asked Questions, a collection of commonly asked questions and their corresponding answers.
PDF	Portable Document Format, a file format used to present documents independently of software or hardware.
API	Application Programming Interface, a set of protocols used for building and integrating application software.
UI	User Interface, the visual elements through which a user interacts with the system.
CRUD	Create, Read, Update, Delete, the four fundamental operations of persistent data storage.
HTTPS	Hypertext Transfer Protocol Secure, an encrypted version of HTTP used for secure data transmission.
TLS	Transport Layer Security, a cryptographic protocol used to secure communications over a network.
SMTP	Simple Mail Transfer Protocol, a standard protocol used for sending emails.
REST	Representational State Transfer, an architectural style for web service APIs.
STOMP	Simple Text Oriented Messaging Protocol, used over WebSocket for real-time messaging.
Stripe	Third-party payment service provider used to process course payments through Payment Intents and webhooks.
Payment Intent	A Stripe object that tracks the lifecycle of a single payment attempt.
Webhook	An HTTP callback by which an external provider notifies the system of an event, such as a payment outcome.
Kafka	A distributed event-streaming platform used for asynchronous communication between services.
Redis	An in-memory data store used for caching and distributed coordination.

Term	Definition
Flyway	A database migration tool used to version and apply database schema changes.
Eureka	A service registry used by microservices for service discovery.
S3-compatible storage	Object storage that exposes the Amazon S3 API, used to store uploaded file content.
Microservice	An independently deployable service that implements a single business domain.

1.4 Document Overview

This document is organised into multiple chapters that collectively define the Software Requirement Specification (SRS) for the Mentify Digital Learning Platform. Chapter 1 provides the introductory information, including the purpose, scope, definitions, and overall document structure. Chapter 2 describes the project scope in detail and identifies the key stakeholders involved in the system.

Chapter 3 presents the functional requirements of the system, including requirements for the modules added during development, while Chapter 4 specifies the non-functional requirements aligned with recognised quality attributes. Chapter 5 outlines the applicable business rules, system constraints, and underlying assumptions. Chapters 6 and 7 include the use case diagrams, detailed use case descriptions, and activity diagrams that illustrate system interactions and workflows.

Chapter 8 describes the external interface requirements, including user, hardware, software, and communication interfaces. Chapter 9 defines the system's data requirements. Chapter 10 discusses alternative solutions considered during the system design phase. Chapter 11 presents the feasibility study covering technical, economic, operational, and schedule feasibility aspects. Chapter 12 provides a detailed risk analysis. Chapter 13 lists all references used in the preparation of this document. Appendix A records every change made to the original requirements, and Appendix B maps requirement groups to implementation evidence.

Chapter 2: Project Scope and Stakeholders

2.1 Project Scope

The Mentify Digital Learning Platform aims to transform TIT Education's existing manual and static educational processes into a modern, AI-enhanced, student-centred digital learning environment. The platform shall serve four categories of application users: Students, Teachers, Administrators, and Super Administrators. An integrated AI service operates as a supporting subsystem and external service providers (Stripe, AI providers, object storage, and email delivery) support payments, AI functionality, file storage, and notifications. These four roles correspond to the roles enforced by the implemented system.

The system shall support the complete academic workflow of the institution: grades and courses are defined by administrators, assigned to teachers, and offered to students; students pay applicable course fees and request enrolment, which is approved by authorised staff; teachers deliver content and manage assignments, quizzes, examinations, and attendance; and administrators monitor operations through payments, enrolment, and analytics views. Assessment processes are automated where the implemented capability is deterministic, such as the scoring of single-answer multiple-choice quizzes, while assignment evaluation and feedback remain teacher-led.

Furthermore, the platform shall provide a centralised communication environment that connects students and educators through course-based group messaging with real-time delivery, event-driven notifications, an academic calendar, and analytics-driven performance insights. AI assistance is provided through summarisation of lessons and materials, AI-assisted quiz draft generation, and role-authorised conversational analysis for staff. Although the system is initially developed for TIT Education, it is designed with scalability considerations to support future expansion to the broader student community.

2.2 Stakeholders

Stakeholder	Role	Interest / Responsibility
TIT Education (Client)	Project sponsor & primary beneficiary	Approves requirements, provides domain knowledge, and funds the project
Students	Primary end users	Use the platform for learning, assessments, payments, communication, and progress tracking
Teachers	Primary end users	Manage assigned courses, deliver content, assess students, record attendance, and monitor performance
Administrators	Internal operators	Manage user accounts, grades, fees, courses, enrolments, payments, materials, marketplace products, and analytics
Super Administrators	Highest administrative authority	Create administrator accounts and perform privileged administrative management
Support Staff (organisational)	Institutional stakeholder	Assist users in practice; no dedicated application role or ticketing portal exists in the current release (deferred)
Development Team (Mentify)	System builders	Design, develop, test, and deploy the platform
Project Supervisor	Academic oversight	Provides guidance to the development team and evaluates project deliverables
External Service Providers	Supporting systems	Stripe (payments), AI providers, S3-compatible object storage, and SMTP email delivery

Chapter 3: Functional Requirements

This chapter specifies the functional requirements of the Mentify Digital Learning Platform. These requirements describe the expected system behaviours and operations for Students, Teachers, Administrators, Super Administrators, and the integrated AI subsystem, followed by requirements for the system-wide modules added during development. Original requirement identifiers have been preserved wherever the requirement remains valid; new requirements have been allocated new identifiers.

The Status column records the implementation status at the time of this revision: Implemented (present in the backend implementation; formal test evidence is recorded in Appendix B as testing progresses), Partially Implemented (part of the requirement is present), Deferred (not part of the current release; retained for traceability), and Not Verified (not established in the reviewed backend implementation and not to be treated as complete until evidence is available).

3.1 Student Functional Requirements

ID	Description	Priority	Source	Status
FR-S001	The system shall allow a new student to be registered by entering their name, email, grade, and password (email/password registration). Google authentication is not supported in the current release.	High	Student, Admin	Implemented
FR-S002	The system shall allow students to log in using JWT-based secure authentication with access and refresh tokens.	High	Student	Implemented
FR-S003	The system shall allow students to view and edit their profile including photo and email.	Medium	Student	Not Verified
FR-S004	The system shall allow students to reset a forgotten password, and to change their account password, using the implemented one-time-password (OTP) verification flow.	Medium	Student	Implemented

ID	Description	Priority	Source	Status
FR-S005	The system shall allow students to log out with secure session termination.	High	Student	Implemented
FR-S006	The system shall allow students or parents to view and update parental contact information.	Medium	Student, Parent	Not Verified
FR-S007	The system shall allow students to view all available courses with filtering by grade level.	High	Student	Implemented
FR-S008	The system shall allow students to request enrolment in a published course. For a paid course, successful payment shall be verified first, and the enrolment shall be created with status PENDING until it is approved by authorised staff.	High	Student	Implemented
FR-S009	The system shall allow students to view a list of all their enrolled courses.	High	Student	Implemented
FR-S010	The system shall allow students to drop (unenrol from) a course. Student-initiated unenrolment is not available in the current release; enrolment removal is performed by authorised staff.	Medium	Student	Deferred
FR-S011	The system shall provide AI-based course recommendations by analysing the student's learning history.	Medium	Student, AI System	Deferred
FR-S012	The system shall display a course dashboard with an overview of modules, lessons, assessments, and progress.	High	Student	Not Verified
FR-S013	The system shall allow students to download study materials associated with their courses. Authorisation shall be checked before a time-limited download link is generated.	High	Student	Implemented
FR-S014	The system shall allow students to take and save personal notes within the platform.	Low	Student	Not Verified
FR-S015	The system shall allow students to search topics by keyword or chapter across their courses.	Medium	Student	Not Verified
FR-S016	The system shall allow students to mark individual lessons as completed.	Medium	Student	Not Verified

ID	Description	Priority	Source	Status
FR-S017	The system shall provide AI summarisation of lessons and permitted materials to students. Conversational AI chat is restricted to Teacher, Admin, and Super Admin roles (see FR-AI003).	Medium	Student, AI System	Partially Implemented
FR-S018	The system shall allow students to view lesson videos and multimedia content.	High	Student	Not Verified
FR-S019	The system shall allow students to bookmark lessons for quick access.	Low	Student	Not Verified
FR-S020	The system shall display all assignments and their due dates to students.	High	Student	Implemented
FR-S021	The system shall allow students to upload and submit completed assignments, subject to the allowed attempts and late-submission rules configured for the assignment.	High	Student	Implemented
FR-S022	The system shall allow students to attempt online quizzes consisting of single-answer multiple-choice (MCQ) questions. Short-answer questions are not supported in the current release.	High	Student	Implemented
FR-S023	The system shall display quiz results including marks, percentage, and pass/fail outcome after completion, according to the quiz result-visibility setting.	High	Student	Implemented
FR-S024	The system shall allow students to view all their grades and scores across assignments and quizzes.	High	Student	Implemented
FR-S025	The system shall allow students to view written feedback provided by the teacher. AI-generated assignment feedback is not part of the current release.	Medium	Student, Teacher	Implemented
FR-S026	The system shall allow students to chat with classmates in group chats within a course.	Medium	Student	Implemented
FR-S027	The system shall allow students to send private messages to teachers.	Medium	Student, Teacher	Not Verified
FR-S028	The system shall allow students to participate in topic-based discussion forums within a course.	Medium	Student	Not Verified
FR-S029	The system shall provide a 24/7 AI-powered help chatbot for study questions, platform issues, and course content, with escalation to support tickets.	Medium	Student, AI System	Deferred

ID	Description	Priority	Source	Status
FR-S030	The system shall deliver notifications to students about supported events such as assignment grading, enrolment status, and messages.	High	Student	Implemented
FR-S031	The system shall display progress and performance information showing scores and learning progress.	Medium	Student	Partially Implemented
FR-S032	The system shall allow students to set personal learning goals.	Low	Student	Not Verified
FR-S033	The system shall allow students to view their attendance records and summaries.	Medium	Student	Implemented
FR-S034	The system shall allow students to initiate payment for a paid course and view the status of their own payments.	High	Student	Implemented
FR-S035	The system shall allow students to view their own payment history.	Medium	Student	Implemented
FR-S036	The system shall allow students to view upcoming examinations and their own examination results, subject to publication rules.	High	Student	Implemented
FR-S037	The system shall allow students to view academic calendar events relevant to them.	Medium	Student	Implemented
FR-S038	The system shall allow students and public visitors to browse the published educational material marketplace catalogue.	Low	Student, Public	Implemented

3.2 Teacher Functional Requirements

ID	Description	Priority	Source	Status
FR-T001	The system shall allow a new teacher account to be registered by an administrator, capturing personal details, subject specialisation, and credentials.	High	Teacher, Admin	Implemented
FR-T002	The system shall allow teachers to log in using JWT-based secure authentication.	High	Teacher	Implemented
FR-T003	The system shall allow teachers to view and edit their profile information.	Medium	Teacher	Not Verified
FR-T004	The system shall allow teachers to reset their password via the OTP-based “forgot password” mechanism.	Medium	Teacher	Implemented

ID	Description	Priority	Source	Status
FR-T005	The system shall allow teachers to log out with secure session termination.	High	Teacher	Implemented
FR-T006	The system shall allow authorised teachers to create a course with name, course code, description, grade, subject, and assigned teacher, together with optional attributes (see FR-GC003 to FR-GC006).	High	Teacher	Implemented
FR-T007	The system shall allow teachers to edit courses assigned to them. Deletion of a course is restricted to Admin and Super Admin roles.	High	Teacher	Implemented
FR-T008	The system shall allow teachers to view the list of students enrolled in each of their courses.	High	Teacher	Implemented
FR-T009	The system shall allow teachers to approve or reject (update the status of) student enrolment requests for courses assigned to them.	Medium	Teacher	Implemented
FR-T010	The system shall allow teachers to update course contents and materials.	High	Teacher	Implemented
FR-T011	The system shall allow teachers to set scheduled time slots for class sessions through the course schedule and calendar events.	Medium	Teacher	Partially Implemented
FR-T012	The system shall allow teachers to manage and update their availability schedule.	Medium	Teacher	Not Verified
FR-T013	The system shall allow teachers to upload learning materials including PDFs, videos, and slides, validated against configured file rules.	High	Teacher	Implemented
FR-T014	The system shall allow teachers to organise lessons into modules, chapters, or units.	High	Teacher	Implemented
FR-T015	The system shall allow teachers to edit or delete individual lessons.	Medium	Teacher	Not Verified
FR-T016	The system shall allow teachers to schedule lesson release dates.	Medium	Teacher	Not Verified
FR-T017	The system shall allow teachers to view student engagement data including lesson views and study time.	Medium	Teacher	Not Verified
FR-T018	The system shall allow teachers to add subtitles or transcripts to video lessons.	Low	Teacher	Not Verified
FR-T019	The system shall allow teachers to set prerequisite lessons.	Medium	Teacher	Not Verified

ID	Description	Priority	Source	Status
FR-T020	The system shall allow teachers to post course announcements to enrolled students.	Medium	Teacher	Partially Implemented
FR-T021	The system shall allow teachers to create assignments with instructions, start date, due date, maximum marks, allowed attempts, late-submission flag, late penalty percentage, and rubric items.	High	Teacher	Implemented
FR-T022	The system shall allow teachers to view all submitted assignments for their courses.	High	Teacher	Implemented
FR-T023	The system shall allow teachers to grade student assignments by entering marks and feedback.	High	Teacher	Implemented
FR-T024	The system shall allow teachers to create online quizzes with single-answer MCQ questions, duration, pass mark, maximum attempts, start/end time, and result-visibility setting.	High	Teacher	Implemented
FR-T025	The system shall provide AI-assisted generation of draft quiz questions based on course content, which the teacher shall review before publication.	Medium	Teacher, AI System	Implemented
FR-T026	The system shall allow teachers to control the publication of quizzes and the visibility of quiz results to students.	High	Teacher	Implemented
FR-T027	The system shall provide AI-assisted grading for assignments and quizzes. In the current release, MCQ quiz scoring is automatic and deterministic, and assignment grading is performed by teachers; AI-generated grading is deferred.	Medium	Teacher, AI System	Deferred
FR-T028	The system shall allow teachers to set assignment deadlines and submission rules (attempts, late submission, and penalty).	High	Teacher	Implemented
FR-T029	The system shall allow teachers to return graded assignments with written feedback.	High	Teacher	Implemented
FR-T030	The system shall allow teachers to chat with students in course groups.	Medium	Teacher	Implemented
FR-T031	The system shall allow teachers to post class-wide announcements.	Medium	Teacher	Partially Implemented
FR-T032	The system shall allow teachers to moderate discussion forums.	Medium	Teacher	Not Verified

ID	Description	Priority	Source	Status
FR-T033	The system shall allow teachers to send bulk notifications to all students in a course.	Medium	Teacher	Not Verified
FR-T034	The system shall display a student performance view with gradebook, student academic summaries, quiz and assignment analytics, and progress.	High	Teacher	Implemented
FR-T035	The system shall allow teachers to manually trigger notifications and alerts to students.	Medium	Teacher	Not Verified
FR-T036	The system shall provide AI-assisted insights into quiz, assignment, attendance, and student progress through authorised AI tools, available through the AI conversation. Predictive weakness detection is not claimed.	Medium	Teacher, AI System	Partially Implemented
FR-T037	The system shall allow teachers to export student performance reports.	Medium	Teacher	Not Verified
FR-T038	The system shall allow teachers to view attendance reports and attendance analytics.	Medium	Teacher	Implemented
FR-T039	The system shall allow teachers to create attendance sessions and record student attendance (see Section 3.12).	High	Teacher	Implemented
FR-T040	The system shall allow authorised teachers to manage examinations and record examination marks (see Section 3.13).	Medium	Teacher	Implemented
FR-T041	The system shall allow teachers to use the role-authorised AI conversation for academic analysis (see Section 3.5).	Medium	Teacher, AI System	Implemented
FR-T042	The system shall allow teachers to view and manage applicable calendar events.	Medium	Teacher	Implemented

3.3 Administrator and Super Administrator Functional Requirements

ID	Description	Priority	Source	Status
FR-A001	The system shall allow the admin to register and edit teacher and student accounts, resend invitations with temporary credentials, and update account details.	High	Admin	Implemented

ID	Description	Priority	Source	Status
FR-A002	The system shall allow the admin to assign user roles from the fixed role set (Student, Teacher, Admin, Super Admin).	High	Admin	Implemented
FR-A003	The system shall allow the admin to deactivate or reactivate user accounts through account status updates and to delete user accounts.	Medium	Admin	Implemented
FR-A004	The system shall allow the admin to configure Single Sign-On (SSO).	Medium	Admin	Deferred
FR-A005	The system shall allow the admin to configure Multi-Factor Authentication (MFA).	Medium	Admin	Not Verified
FR-A006	The system shall enforce role-based access control (RBAC) on protected operations according to the fixed role model.	High	Admin	Implemented
FR-A007	The system shall allow the admin to create custom roles beyond the default four. Custom role creation is not supported in the current release.	Low	Admin	Deferred
FR-A008	The system shall allow the admin to view all courses and content on the platform.	High	Admin	Implemented
FR-A009	The system shall allow the admin to audit course content for compliance with institutional standards.	Medium	Admin	Not Verified
FR-A010	The system shall allow the admin to publish or unpublish courses.	High	Admin	Implemented
FR-A011	The system shall allow the admin to manage course categories and tags.	Medium	Admin	Not Verified
FR-A012	The system shall allow the admin to assign a teacher to a course.	High	Admin	Implemented
FR-A013	The system shall allow the admin to view usage and institutional analytics (see Section 3.16). Report export is not claimed.	Medium	Admin	Partially Implemented
FR-A014	The system shall allow the admin to analyse system engagement metrics through enrolment, assessment, attendance, and payment analytics.	Medium	Admin	Partially Implemented
FR-A015	The system shall allow the admin to create and manage academic grades.	High	Admin	Implemented
FR-A016	The system shall allow the admin to configure grade-based monthly course fees.	High	Admin	Implemented
FR-A017	The system shall allow the admin to approve or reject student enrolments and monitor enrolment status.	High	Admin	Implemented

ID	Description	Priority	Source	Status
FR-A018	The system shall allow the admin to view, filter, and inspect payment records and payment details.	High	Admin	Implemented
FR-A019	The system shall allow the admin to process supported refunds for payments.	Medium	Admin	Implemented
FR-A020	The system shall allow the admin to upload materials and create, edit, and publish marketplace products (see Section 3.11).	Medium	Admin	Implemented
FR-A021	The system shall allow the admin to manage examinations, attendance, and calendar events within applicable authorisation rules.	Medium	Admin	Implemented
FR-A022	The system shall allow the admin to use the role-authorised AI conversation for institutional analysis.	Low	Admin, AI System	Implemented
FR-SA001	The system shall allow the Super Admin to register administrator accounts.	High	Super Admin	Implemented
FR-SA002	The system shall allow the Super Admin to perform all Administrator functions and privileged administrative operations, including course deletion.	High	Super Admin	Implemented

3.4 Support Team Functional Requirements (Deferred)

The implemented role model does not contain a Support Team role, and no helpdesk ticketing portal or knowledge base has been established. The requirements below are retained from the original document for traceability and are classified as deferred. Support responsibilities in the institution are performed by Administrators using the Administrator functions in Section 3.3.

ID	Description	Priority	Source	Status
FR-SP001	The system shall allow the support team to view and filter support tickets by status, priority, or type.	High	Support Team	Deferred
FR-SP002	The system shall allow the support team to assist users with login and password issues.	High	Support Team	Deferred
FR-SP003	The system shall allow the support team to guide users through platform features.	Medium	Support Team	Deferred
FR-SP004	The system shall allow the support team to troubleshoot technical glitches.	High	Support Team	Deferred

ID	Description	Priority	Source	Status
FR-SP005	The system shall allow the support team to provide course enrolment support.	Medium	Support Team	Deferred
FR-SP006	The system shall allow the support team to create and edit FAQ articles.	Medium	Support Team	Deferred
FR-SP007	The system shall allow the support team to organise knowledge base categories.	Medium	Support Team	Deferred
FR-SP008	The system shall allow the support team to identify gaps in knowledge base content.	Low	Support Team	Deferred
FR-SP009	The system shall allow the support team to escalate complex issues to admins or developers.	High	Support Team	Deferred
FR-SP010	The system shall allow the support team to collaborate with teachers on student issues.	Medium	Support Team	Deferred
FR-SP011	The system shall provide the support team with read-only access to relevant admin tools.	Medium	Support Team	Deferred

3.5 AI System Functional Requirements

AI access is defined at the feature level. Lesson and material summarisation is available to Student, Teacher, Admin, and Super Admin roles. The main conversational AI (REST and WebSocket chat) is available to Teacher, Admin, and Super Admin roles only. Provider availability depends on implemented adapters and runtime configuration.

ID	Description	Priority	Source	Status
FR-AI001	The AI system shall analyse student learning history and provide personalised course recommendations.	Medium	AI System, Student	Deferred
FR-AI002	The AI system shall summarise permitted lesson and material content on demand for authorised users.	Medium	AI System, Student	Implemented
FR-AI003	The AI system shall provide a conversational assistant, through REST and WebSocket interfaces, for Teacher, Admin, and Super Admin roles, answering academic and administrative questions using authorised data tools.	Medium	AI System, Staff	Implemented
FR-AI004	The AI system shall generate draft quiz questions based on course content for teacher review.	Medium	AI System, Teacher	Implemented

ID	Description	Priority	Source	Status
FR-AI005	The AI system shall assist in grading assignments and quizzes. In the current release, MCQ scoring is deterministic and assignment grading is teacher-performed; AI grading is deferred.	Medium	AI System, Teacher	Deferred
FR-AI006	The AI system shall provide analysis of quiz performance, assignment performance, attendance performance, student progress, registration overview, and upcoming calendar events through authorised tools.	Medium	AI System, Teacher	Partially Implemented
FR-AI007	The AI system shall generate feedback on student submissions.	Medium	AI System, Teacher, Student	Deferred
FR-AI008	The system shall persist AI conversations and messages and allow authorised users to continue a conversation using its identifier.	Medium	AI System	Implemented
FR-AI009	The system shall support configurable AI provider integrations, with provider availability dependent on implemented adapters and runtime configuration.	Medium	AI System	Implemented
FR-AI010	The system shall enforce AI usage limits and input validation, and apply configured content filtering and prompt-injection checks.	High	AI System	Implemented
FR-AI011	The system shall maintain AI request audit records and handle provider timeouts, retries, and failures.	Medium	AI System	Implemented
FR-AI012	The system shall enforce role and data-scope restrictions on AI tools so that AI cannot access data beyond the requester's authorisation.	High	AI System	Implemented

3.6 Authentication and Account Management Requirements

ID	Description	Priority	Source	Status
FR-AU001	The authentication service shall authenticate users using registered credentials and issue signed JWT access tokens.	High	All Roles	Implemented

ID	Description	Priority	Source	Status
FR-AU002	The system shall validate access tokens (signature, issuer, and expiry) before permitting protected operations.	High	All Roles	Implemented
FR-AU003	The system shall support refresh-token-based session renewal and logout.	High	All Roles	Implemented
FR-AU004	The system shall expose the public keys required to verify issued tokens (JWKS).	Medium	System	Implemented
FR-AU005	The system shall provide OTP-based password recovery and reset.	Medium	All Roles	Implemented
FR-AU006	The system shall associate authentication identities with user-service profiles, keeping credentials separate from profile data.	High	System	Implemented
FR-AU007	The system shall provision accounts by invitation with temporary credentials and allow invitations to be resent.	High	Admin	Implemented
FR-AU008	The system shall restrict privileged administrative operations, including administrator account creation, to authorised roles.	High	Super Admin	Implemented

3.7 Grade and Course Management Requirements

ID	Description	Priority	Source	Status
FR-GC001	Authorised administrators shall create and manage academic grades.	High	Admin	Implemented
FR-GC002	Authorised administrators shall configure grade-level monthly fees.	High	Admin	Implemented
FR-GC003	Authorised users shall create courses associated with an existing grade; each course shall identify its assigned teacher.	High	Admin, Teacher	Implemented
FR-GC004	A course record shall contain course name, course code, description, subject, and optional thumbnail.	High	Admin, Teacher	Implemented
FR-GC005	A course may carry optional academic year, semester, level, capacity, credits, schedule, grading weights, and teaching assistant attributes.	Medium	Admin, Teacher	Implemented
FR-GC006	A course shall support online, visible, and published flags to control its availability.	Medium	Admin, Teacher	Implemented

ID	Description	Priority	Source	Status
FR-GC007	The system shall organise course content as course, module, and lesson, with module objectives, lesson content, duration, and publication state.	High	Teacher	Partially Implemented
FR-GC008	The system shall enforce role-based restrictions on course modification and deletion; deletion is restricted to Admin and Super Admin.	High	Admin, Super Admin	Implemented

3.8 Enrolment Management Requirements

ID	Description	Priority	Source	Status
FR-EN001	Students shall be able to request enrolment in available courses; the system shall validate the course identifier.	High	Student	Implemented
FR-EN002	The system shall prevent duplicate active enrolment in the same course.	High	Student	Implemented
FR-EN003	The system shall verify successful payment before student self-enrolment in a paid course.	High	Student	Implemented
FR-EN004	Student self-enrolment requests shall initially be recorded with status PENDING; supported statuses are PENDING, APPROVED, and REJECTED.	High	Student	Implemented
FR-EN005	Authorised teachers and administrators shall manage enrolment status; teachers shall manage only enrolments of courses assigned to them.	High	Teacher, Admin	Implemented
FR-EN006	Administrators and teachers shall be able to create enrolments directly through the staff-managed path, which is subject to separate authorisation checks.	Medium	Teacher, Admin	Implemented
FR-EN007	The system shall synchronise course communication-group membership with enrolment, using the implemented strict and non-strict handling paths.	Medium	System	Implemented
FR-EN008	The system shall provide enrolment statistics and course popularity information.	Medium	Admin	Implemented

3.9 Payment Management Requirements

ID	Description	Priority	Source	Status
FR-PY001	The system shall identify payable courses using the applicable grade and course fee information.	High	System	Implemented
FR-PY002	Students shall be able to initiate supported course payments; the system shall create a Stripe Payment Intent for each payable transaction.	High	Student	Implemented
FR-PY003	The system shall maintain payment records and transaction identifiers with status values including pending, success, failed, and refunded.	High	System	Implemented
FR-PY004	The system shall process Stripe webhook events only after verifying their signature, and shall prevent duplicate processing of previously handled events.	High	System	Implemented
FR-PY005	Students shall be able to view their own payment history and retrieve the status of their own payments only.	High	Student	Implemented
FR-PY006	Authorised administrators shall view and filter payment records and payment details.	High	Admin	Implemented
FR-PY007	Authorised administrators shall manage supported refunds.	Medium	Admin	Implemented
FR-PY008	The enrolment service shall verify payment success when required for student self-enrolment.	High	System	Implemented
FR-PY009	The system shall not claim recurring subscription billing; monthly fee values are configuration data used to determine course payment amounts.	Medium	System	Implemented

3.10 Material Management Requirements

ID	Description	Priority	Source	Status
FR-MT001	Authorised users shall upload supported educational files; the system shall validate files against configured rules.	High	Teacher, Admin	Implemented

ID	Description	Priority	Source	Status
FR-MT002	The system shall store file content in configured S3-compatible object storage and maintain metadata and resource associations in the material service.	High	System	Implemented
FR-MT003	The system shall record each material purpose (for example learning material, assignment, submission, examination, AI source, announcement, communication, or marketplace attachment).	Medium	System	Implemented
FR-MT004	The system shall enforce authorisation before issuing a time-limited download URL.	High	All Roles	Implemented
FR-MT005	The system shall support material search and filtering.	Medium	Teacher, Admin	Implemented
FR-MT006	Authorised users shall update metadata, archive, or delete materials according to permission rules.	Medium	Teacher, Admin	Implemented
FR-MT007	The system shall support association of materials with courses, modules, and lessons, and approved AI-source materials.	Medium	Teacher	Implemented
FR-MT008	Administrators shall upload materials intended for marketplace products.	Medium	Admin	Implemented
FR-MT009	The system shall maintain audit information for material operations.	Medium	System	Implemented

3.11 Marketplace Catalogue Requirements

Only the administrator-managed product catalogue is implemented. A complete paid-material checkout, purchase history, and purchaser entitlement workflow is not claimed and is classified as deferred (FR-MK007).

ID	Description	Priority	Source	Status
FR-MK001	Administrators shall create educational material product records referencing uploaded material resources.	Medium	Admin	Implemented
FR-MK002	Administrators shall manage product title, description, category, subject, grade, and thumbnail.	Medium	Admin	Implemented

ID	Description	Priority	Source	Status
FR-MK003	Product categories shall include notes, short notes, past papers, exam papers, model papers, worksheets, question banks, marking schemes, revision packs, study guides, tutorials, practice papers, and other resources.	Low	Admin	Implemented
FR-MK004	Products shall support free or paid pricing, a discount percentage, and currency; the system shall calculate the final price.	Medium	Admin	Implemented
FR-MK005	Administrators shall manage product publication status.	Medium	Admin	Implemented
FR-MK006	Public users shall be able to browse and search published products only.	Medium	Public	Implemented
FR-MK007	The system shall provide marketplace checkout, purchase history, and purchaser access to paid materials.	Low	Student	Deferred

3.12 Attendance Management Requirements

ID	Description	Priority	Source	Status
FR-AT001	Authorised users shall create attendance sessions associated with relevant academic activities.	High	Teacher, Admin	Implemented
FR-AT002	Authorised teachers shall record student attendance by session.	High	Teacher	Implemented
FR-AT003	The system shall maintain attendance records by student and session and support session status management and completion.	High	System	Implemented
FR-AT004	The system shall calculate attendance summaries and provide attendance analytics to authorised users.	Medium	Teacher, Admin	Implemented
FR-AT005	The system shall support calendar synchronisation for attendance sessions.	Low	System	Implemented

3.13 Examination Management Requirements

ID	Description	Priority	Source	Status
FR-EX001	Authorised users shall create and manage examination records associated with academic courses; types shall include MIDTERM, FINAL, PRACTICAL, and OTHER.	High	Teacher, Admin	Implemented
FR-EX002	The system shall support examination scheduling, instructions, and attachments.	Medium	Teacher, Admin	Implemented
FR-EX003	Authorised users shall record examination marks and maintain result sheets.	High	Teacher, Admin	Implemented
FR-EX004	Students shall be able to view upcoming examinations and their own results, subject to authorisation and publication rules.	High	Student	Implemented
FR-EX005	The system shall provide examination analytics.	Medium	Admin, Teacher	Implemented
FR-EX006	The system shall synchronise examination scheduling with the calendar service.	Medium	System	Implemented

3.14 Calendar Management Requirements

ID	Description	Priority	Source	Status
FR-CA001	Authorised users shall create, cancel, and manage calendar events, including venue information.	Medium	Teacher, Admin	Implemented
FR-CA002	The system shall provide calendar queries and filtering appropriate to each role.	Medium	All Roles	Implemented
FR-CA003	The system shall maintain calendar projections for examinations, assignments, and quizzes, and support reconciliation of projections.	Medium	System	Implemented
FR-CA004	The system shall enforce role-based calendar visibility.	Medium	System	Implemented

3.15 Communication and Notification Requirements

ID	Description	Priority	Source	Status
FR-CN001	Authorised users shall communicate through course and other communication groups with real-time delivery over WebSocket.	Medium	All Roles	Implemented
FR-CN002	The system shall maintain communication groups, group membership, and communication settings, and enforce group access restrictions.	Medium	System	Implemented
FR-CN003	The system shall support permitted message attachments.	Low	All Roles	Implemented
FR-CN004	The system shall create notifications for supported application events, such as assignment grading and material operations.	High	System	Implemented
FR-CN005	Users shall be able to access their permitted notifications, and the system shall support the implemented email delivery channel through SMTP.	Medium	All Roles	Partially Implemented

3.16 Analytics and Reporting Requirements

ID	Description	Priority	Source	Status
FR-AN001	Authorised administrators shall access administrative analytics, including user registration statistics.	Medium	Admin	Implemented
FR-AN002	The system shall provide enrolment analytics and course popularity information.	Medium	Admin	Implemented
FR-AN003	The system shall provide assignment, quiz, examination, and attendance analytics to authorised roles.	Medium	Teacher, Admin	Implemented
FR-AN004	The system shall provide payment analytics to authorised administrators.	Medium	Admin	Implemented
FR-AN005	Authorised teachers shall access course and student academic summaries and the gradebook.	High	Teacher	Implemented
FR-AN006	The system shall enforce role and academic-scope restrictions on analytics data.	High	System	Implemented
FR-AN007	The system shall provide downloadable or scheduled formal reports (PDF, Excel, scheduled, or AI-generated institutional reports).	Low	Admin, Teacher	Not Verified

Chapter 4: Non-Functional Requirements

Non-functional requirements define the quality attributes, performance expectations, operational constraints, and system characteristics that the Mentify Digital Learning Platform must satisfy. The numeric values below are requirements and targets. They are not claims of achieved results; achievement shall be recorded only when supported by load-test, monitoring, recovery-test, or security-assessment evidence.

4.1 Performance

ID	Category	Description	Acceptance Criterion
NFR-P001	Performance	The system shall respond to user interactions promptly under normal load conditions.	95% of page load requests shall complete within 2 seconds under normal load (target: up to 500 concurrent users), verified by load testing.
NFR-P002	Performance	The system shall handle search queries efficiently.	95% of search queries shall return results within 2 seconds.
NFR-P003	Performance	The system shall process quiz submissions without delay.	Quiz submission and MCQ result calculation shall complete within 3 seconds.
NFR-P004	Performance	The system shall support concurrent usage during peak hours.	The system shall support a target of 1,000 concurrent users without response time degradation exceeding 20%, verified by performance testing.
NFR-P005	Performance	AI-dependent operations shall have separate latency and timeout limits.	AI requests shall enforce configured provider timeouts and shall not block non-AI operations such as course browsing.
NFR-P006	Performance	File uploads shall be bounded.	The system shall enforce a configured maximum file size and supported file types for uploads.

4.2 Reliability

ID	Category	Description	Acceptance Criterion
NFR-R001	Reliability	The system shall maintain high availability.	Target of 99.5% uptime measured monthly, excluding scheduled maintenance, verified by monitoring.
NFR-R002	Reliability	The system shall recover from failures with minimal data loss.	Target recovery within 30 minutes with no more than 5 minutes of transactional data loss, verified by recovery testing.
NFR-R003	Reliability	The system shall perform automated backups.	Automated database backups shall be scheduled every 24 hours and stored in a geographically separate location.
NFR-R004	Reliability	The system shall handle service and provider failures gracefully.	Services shall expose health endpoints and apply timeouts, retries, and error responses for inter-service and external calls (Stripe, AI providers, storage, email).
NFR-R005	Reliability	The system shall process external events idempotently.	Duplicate payment webhook events shall not cause duplicate status transitions.

4.3 Usability

ID	Category	Description	Acceptance Criterion
NFR-U001	Usability	The system shall be easy to learn for new users.	A new student shall enrol in a course and access lesson content within 5 minutes of first login without assistance.
NFR-U002	Usability	The system shall provide a responsive user interface.	The UI shall function correctly on screens ranging from 320px to 1920px.
NFR-U003	Usability	The system shall meet accessibility standards.	The system shall comply with WCAG 2.1 Level AA guidelines.
NFR-U004	Usability	The system shall support multiple languages.	The system shall support English and Tamil as primary interface languages (target; not verified in the backend).

4.4 Security

ID	Category	Description	Acceptance Criterion
NFR-SE001	Security	The system shall authenticate users securely.	Authentication shall use signed JWT access tokens with a maximum expiry of 24 hours, refresh-token support, and publication of verification keys through JWKS.
NFR-SE002	Security	The system shall encrypt sensitive data.	Data in transit shall use TLS 1.2 or higher; passwords shall be hashed using bcrypt with a minimum cost factor of 12 (aligned with the OWASP checklist of the development guidelines) and stored only by the authentication service.
NFR-SE003	Security	The system shall enforce role-based access control.	Users shall not access functionality beyond assigned permissions; authorisation shall be enforced at endpoint level and, where applicable, resource-ownership level, independently of frontend route visibility.
NFR-SE004	Security	The system shall protect against common web vulnerabilities.	The system shall pass an OWASP Top 10 assessment with no critical or high-severity findings. Role-based access alone does not demonstrate OWASP compliance.
NFR-SE005	Security	The system shall support Multi-Factor Authentication.	MFA is deferred in the current release; it is retained as a target for administrator accounts.
NFR-SE006	Security	The system shall protect payment operations.	Stripe webhook events shall be signature-verified before payment status transitions; students shall access only their own payment records.
NFR-SE007	Security	The system shall protect stored files.	Material access shall be authorised before a time-limited download URL is issued; ownership and visibility rules shall be applied.
NFR-SE008	Security	The system shall protect AI operations.	AI requests shall be subject to usage limits, content filtering, and prompt-injection checks; AI tools shall enforce the requester's role and data scope; sensitive data shall not be exposed beyond authorisation.
NFR-SE009	Security	The system shall protect configuration secrets.	Sensitive configuration values shall be supplied through secure runtime configuration and not stored in source control.

4.5 Scalability

ID	Category	Description	Acceptance Criterion
NFR-SC001	Scalability	The system shall scale to accommodate growth.	The architecture shall allow domain services to be deployed and scaled independently, using Redis-supported caching and coordination. Scale targets (for example a 300% user increase) shall be validated by load testing.
NFR-SC002	Scalability	The system shall handle growing data volumes.	Target of up to 1 million course records and 10 million activity records with performance degradation not exceeding 10%, verified by database profiling.

4.6 Maintainability

ID	Category	Description	Acceptance Criterion
NFR-M001	Maintainability	The system shall follow modular architecture principles.	The system shall adopt a microservice architecture (Java 17, Spring Boot 3.2.5) composed of 14 domain services, an API gateway, a configuration server, and a service registry, enabling independent module deployment.
NFR-M002	Maintainability	The system shall maintain comprehensive code documentation and testing.	APIs shall be documented (OpenAPI/Swagger); automated test coverage target shall exceed 80%.
NFR-M003	Maintainability	The system shall support continuous integration and deployment.	Automated CI/CD configuration shall build and test the services; automated deployment shall complete within 30 minutes. Pipeline success shall be recorded as achieved only with execution evidence.
NFR-M004	Maintainability	The system shall manage schema and configuration changes in a controlled way.	Database schema changes shall be applied through versioned Flyway migrations and service configuration shall be centralised in the Config Server.

4.7 Portability

ID	Category	Description	Acceptance Criterion
NFR-PO001	Portability	The system shall be accessible across major web browsers.	The system shall function on the latest two versions of Chrome, Firefox, Safari, and Edge.
NFR-PO002	Portability	The system shall be deployable across different hosting environments.	The system shall be containerised using Docker and Docker Compose, with a Caddy reverse-proxy configuration. AWS (EC2) is the intended deployment environment; Azure and Kubernetes deployment are targets not yet verified.

Chapter 5: Business Rules, Constraints and Assumptions

This chapter defines the governing business policies, operational limitations, and underlying assumptions that influence the design, development, and implementation of the Mentify Digital Learning Platform.

5.1 Business Rules

ID	Business Rule
BR-001	A student may only enrol in courses that match their registered grade level unless explicitly approved by a teacher or administrator.
BR-002	Each course must be associated with a valid academic grade and have an assigned teacher before it can be published and made visible to students.
BR-003	Assignment submissions shall not be accepted after the deadline unless the teacher has enabled a late submission policy for that assignment; submission attempts shall not exceed the allowed attempts, and a configured late penalty (0–100%) shall be applied.
BR-004	A student must complete all prerequisite lessons before accessing lessons configured with prerequisites by the teacher (applicable once prerequisite configuration is verified; see FR-T019).
BR-005	Only users with the Administrator or Super Administrator role may register, modify, or deactivate user accounts. Only the Super Administrator may create administrator accounts. Teachers shall not manage accounts of other teachers.
BR-006	The application role model is fixed to Student, Teacher, Admin, and Super Admin. No Support Team role exists in the current release; support duties are performed by Administrators.
BR-007	AI-generated quiz questions are drafts and must be reviewed and approved by the teacher before the quiz is published to students.
BR-008	All communication within the platform, including group chats and messages, shall comply with TIT Education's code of conduct and be subject to moderation.
BR-009	Notifications shall be created for supported events; delivery shall use the implemented channels, including email through SMTP where configured.
BR-010	A student shall not self-enrol in a paid course without successful payment verification.
BR-011	Student self-enrolment shall initially have status PENDING; supported statuses are PENDING, APPROVED, and REJECTED.

ID	Business Rule
BR-012	Enrolment status may be updated only by authorised teachers (for their assigned courses) and administrators.
BR-013	Grade-level monthly fees shall be maintained only by authorised administrators.
BR-014	Only Administrators and Super Administrators may delete courses.
BR-015	Users shall access protected materials only when authorised by their enrolment, ownership, or other applicable visibility policy.
BR-016	Marketplace products shall be created and published only by authorised administrators; products marked free shall have a zero price, paid products shall have a positive base price, and only published products shall appear in the public catalogue.
BR-017	Quiz question types in the current release are limited to single-answer multiple-choice questions, and MCQ marks shall be calculated from stored correct answers.
BR-018	Teacher grading operations shall be restricted to assignments of courses assigned to the teacher.
BR-019	Only authorised users shall access AI conversation histories and academic analytics; AI requests shall comply with configured usage and content restrictions.
BR-020	Payment status changes shall follow verified provider events and idempotent processing; students shall access only their own payment records.
BR-021	Communication-group membership shall be managed according to the authorised course membership flow.

5.2 Constraints

ID	Constraint
C-001	The system must be developed within the academic timeline defined by the SENG 31242 module and the SENG 34213 System Development Project.
C-002	The development team consists of four members (Holins, Dilakshan, Jenushan, Zaina), which limits the feature scope deliverable in the initial release.
C-003	The system shall be accessible via standard web browsers and shall not require native mobile applications in the initial release.
C-004	AI functionalities are dependent on the availability, performance, and cost of third-party AI/ML service providers.
C-005	The system must comply with Sri Lankan data protection regulations and institutional policies of TIT Education.

ID	Constraint
C-006	Initial deployment must align with the existing infrastructure capabilities of TIT Education.
C-007	This revision describes the backend implementation; frontend completion and production deployment status require separate verification, and source code is not changed by documentation updates.
C-008	Course payments depend on the availability of Stripe; marketplace checkout depends on future development.

5.3 Assumptions

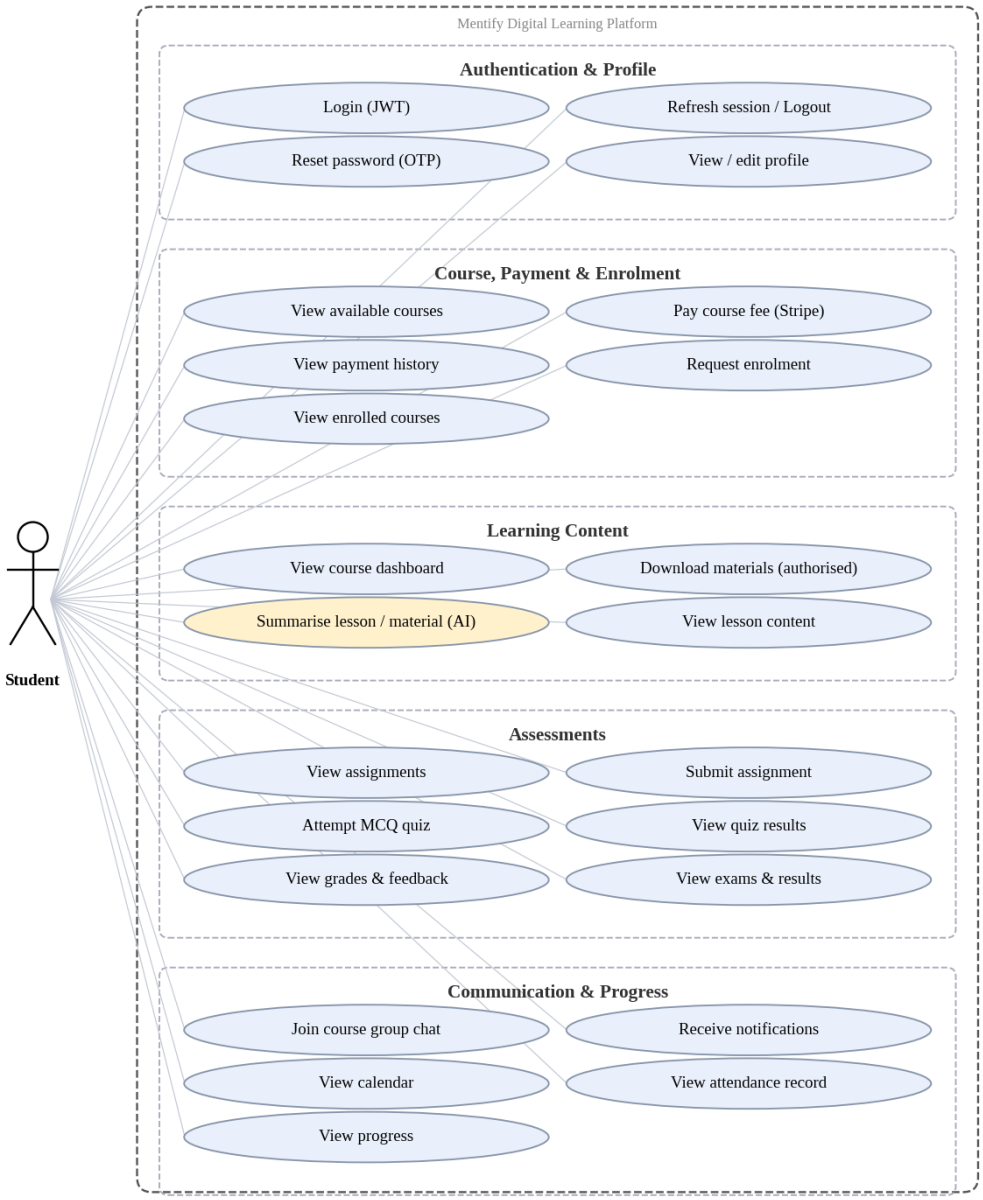
ID	Assumption
A-001	It is assumed that all students and teachers have access to internet-enabled devices such as laptops, tablets, or smartphones.
A-002	It is assumed that TIT Education will provide the required domain expertise, user feedback, and initial course content.
A-003	It is assumed that users possess basic digital literacy required to navigate a web-based application.
A-004	It is assumed that TIT Education will designate at least one staff member to function as the system administrator.
A-005	It is assumed that the institution will provide or procure the necessary cloud hosting infrastructure for deployment.
A-006	It is assumed that continuous internet connectivity is required for all system operations.
A-007	It is assumed that PostgreSQL, object storage, Stripe, SMTP, and configured AI providers are available when dependent features are used. Stripe unavailability may prevent payment initiation; an AI provider outage prevents AI features but does not prevent course browsing.
A-008	It is assumed that AWS EC2 is the intended deployment environment; Docker Compose and production configuration demonstrate deployment preparation, not a verified production deployment.

Chapter 6: Use Case Diagrams and Descriptions

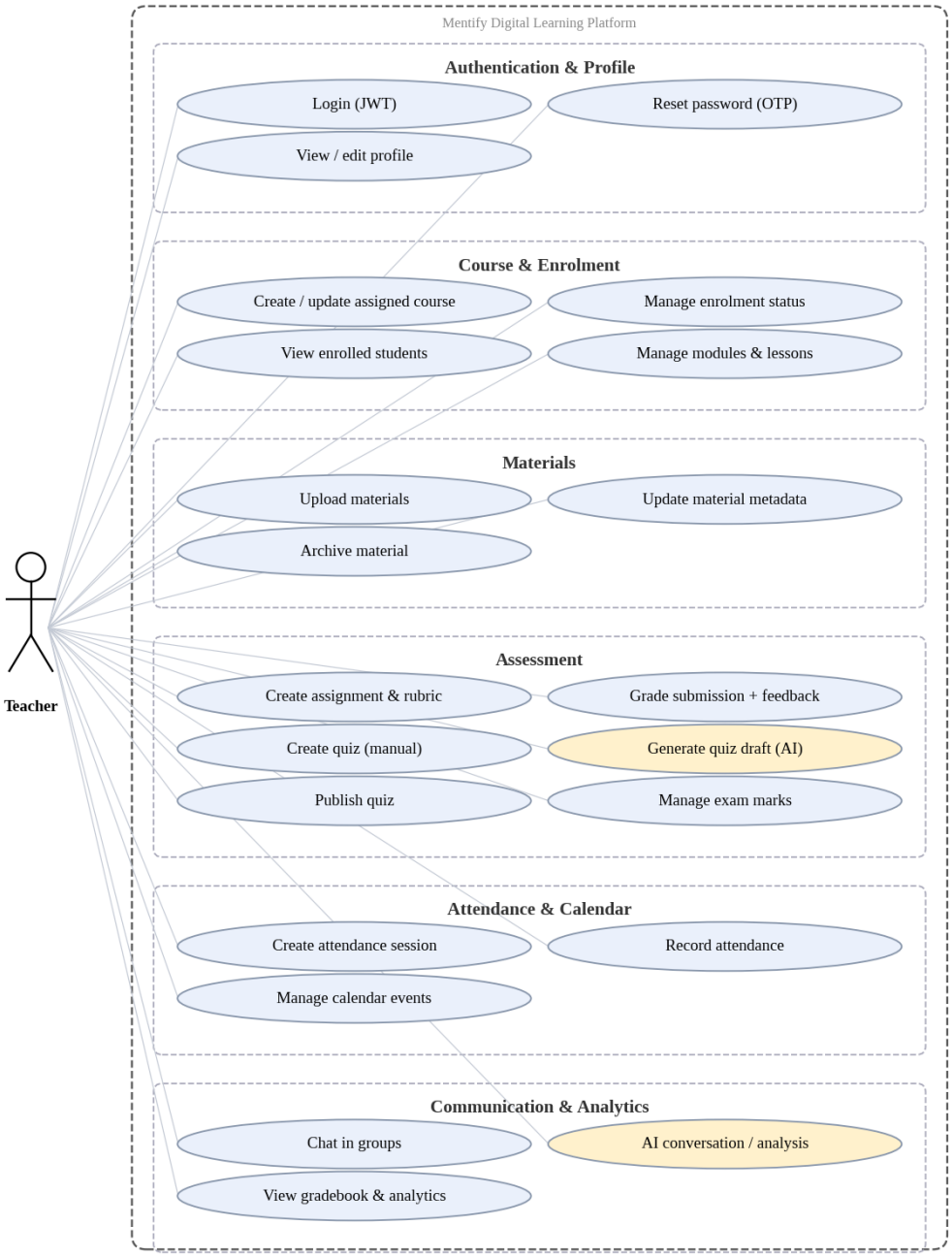
6.1 Use Case Diagrams

The use case diagrams illustrate the interactions between the primary actors and the Mentify Digital Learning Platform. The actors correspond to the four implemented roles. External systems (Stripe, AI providers, and object storage) and infrastructure components (Kafka, PostgreSQL, Eureka) are not modelled as human actors; they are shown in the architecture diagram in Chapter 8. AI-involved use cases are highlighted. The following use case diagrams are included in this chapter.

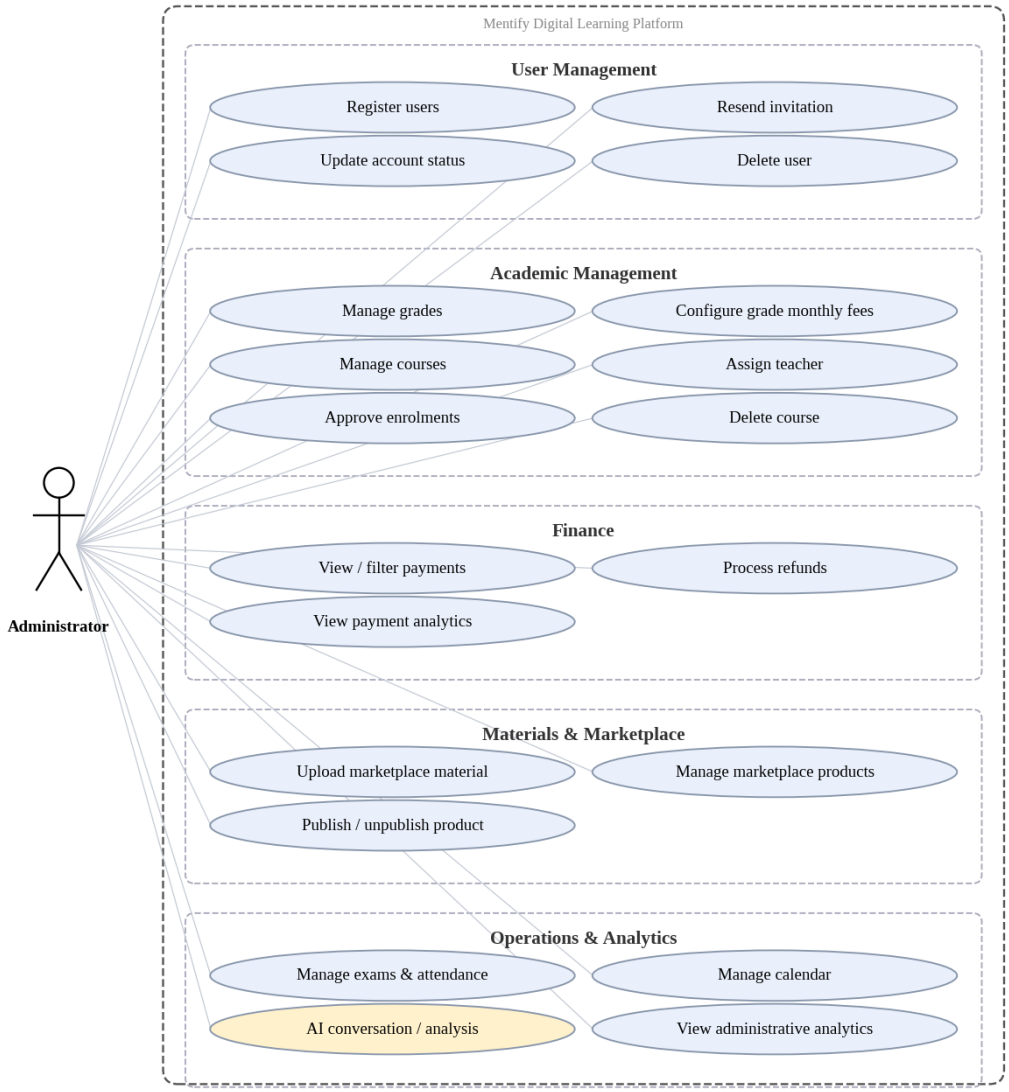
6.1.1 Student Use Case Diagram



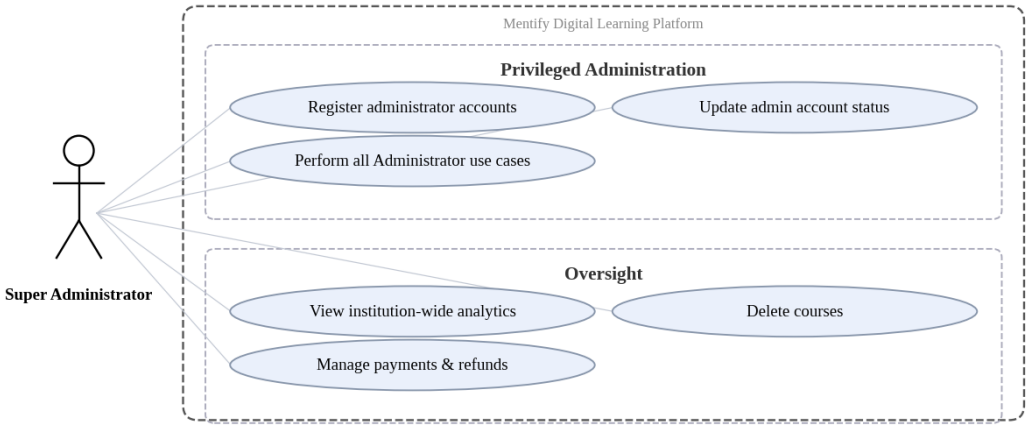
6.1.2 Teacher Use Case Diagram



6.1.3 Administrator Use Case Diagram



6.1.4 Super Administrator Use Case Diagram



The Support Team use case diagram of the original document has been removed from the delivered actor model because no Support Team role exists in the implementation. The corresponding requirements are retained as deferred in Section 3.4.

6.2 Use Case Descriptions

The following use case descriptions present the major system interactions for the primary actors of the Mentify Digital Learning Platform.

Use Case Description: UC-S001

Field	Description
Use Case ID	UC-S001
Use Case Name	Register as Student
Primary Actor	Administrator (on behalf of Student)
Description	A student account is registered through the user management function and the student receives an invitation with temporary credentials.
Preconditions	The administrator is logged in with Admin or Super Admin role.; The email address is not already registered.
Postconditions	A new student account and profile are created and the student is invited to sign in.
Main Flow	1. The administrator opens the user management page.; 2. The system displays the registration form.; 3. The administrator enters the student's name, email, grade level, and role.; 4. The administrator submits the form.; 5. The system validates the input data.; 6. The system creates the account and profile and sends an invitation email with temporary credentials.; 7. The student signs in and changes the temporary password.
Alternative Flow	6a. The administrator selects "Resend invitation" if the student did not receive the email.
Exception Flow	5a. If the email is already registered, the system displays the message: "An account with this email already exists."; 5b. If required fields are missing, the system highlights the missing fields and prompts for completion.
Priority	High

Use Case Description: UC-S002

Field	Description
Use Case ID	UC-S002
Use Case Name	Login (JWT-based)
Primary Actor	Student

Field	Description
Description	The student logs into the system using email and password with JWT-based authentication.
Preconditions	The student has a registered and active account.
Postconditions	The student is authenticated and signed access and refresh tokens are issued for the session.
Main Flow	1. The student navigates to the login page.; 2. The student enters their email and password.; 3. The system validates the credentials.; 4. The system generates a JWT access token and refresh token.; 5. The system redirects the student to their dashboard.
Alternative Flow	2a. The student selects “Forgot password.” The system sends a one-time password (OTP) by email and the student sets a new password.
Exception Flow	3a. If the credentials are invalid, the system displays the message: “Invalid email or password.”; 3b. If the account is deactivated, the system displays the message: “Your account has been deactivated. Please contact an administrator.”
Priority	High

Use Case Description: UC-S008

Field	Description
Use Case ID	UC-S008
Use Case Name	Enrol in Course
Primary Actor	Student
Description	The student selects a course from the available courses list and requests enrolment to gain access to the course content.
Preconditions	The student is logged in.; The course is published and matches the student’s grade level.
Postconditions	An enrolment record exists with status PENDING (and APPROVED once authorised staff approve it).
Main Flow	1. The student navigates to the available courses page.; 2. The student selects a course.; 3. The system displays the course details.; 4. The student clicks “Enrol.”; 5. The system verifies the course and checks for duplicate enrolment.; 6. If the course is paid, the system verifies that a successful payment exists.; 7. The system creates the enrolment with status PENDING.; 8. An authorised teacher or administrator approves the request and the course appears in the student’s enrolled courses list.
Alternative Flow	6a. If payment has not been made, the system rejects the request and directs the student to pay (UC-S009).; 8a. If the request is rejected, the status is set to REJECTED and the student is notified.
Exception Flow	5a. If the student is already enrolled, the system displays the message: “You are already enrolled in this course.”
Priority	High

Use Case Description: UC-S009

Field	Description
Use Case ID	UC-S009
Use Case Name	Pay Course Fee
Primary Actor	Student
Description	The student pays the fee of a paid course through Stripe.
Preconditions	The student is logged in.; The course has a payable fee.
Postconditions	A payment record with status SUCCESS exists and can be used to enrol.
Main Flow	1. The student selects a paid course.; 2. The system resolves the course fee.; 3. The payment service creates a Stripe Payment Intent and a payment record.; 4. The student completes the payment with Stripe.; 5. Stripe sends a webhook event; the system verifies its signature.; 6. The system updates the payment status.; 7. The student views the payment in their payment history.
Alternative Flow	5a. A duplicate webhook event is ignored.
Exception Flow	4a. If the payment fails, the status is set to FAILED and the student may retry.; 3a. If Stripe is unavailable, payment initiation fails with an error message.
Priority	High

Use Case Description: UC-T001

Field	Description
Use Case ID	UC-T001
Use Case Name	Grade Assignment Submission
Primary Actor	Teacher
Description	The teacher evaluates a student’s submission, enters marks and feedback, and returns the result.
Preconditions	The teacher is logged in and assigned to the course.; A submission exists.
Postconditions	The submission is graded or returned for revision and the student is notified.
Main Flow	1. The teacher opens the submissions of an assignment.; 2. The system verifies that the teacher is authorised for the course.; 3. The teacher reviews the submission against the rubric.; 4. The teacher enters marks and written feedback.; 5. The system records the grade and notifies the student.
Alternative Flow	4a. The teacher returns the submission for revision.
Exception Flow	2a. If the teacher is not authorised, the system rejects the request.
Priority	High

Use Case Description: UC-A001

Field	Description
Use Case ID	UC-A001
Use Case Name	Register Administrator Account
Primary Actor	Super Administrator
Description	The Super Administrator registers a new administrator account.
Preconditions	The user is logged in as Super Admin.
Postconditions	A new administrator account is created and an invitation is sent.
Main Flow	1. The Super Admin opens administrator registration.; 2. The Super Admin enters the administrator's details.; 3. The system verifies the caller's role, validates the input, and creates the account.; 4. The system sends an invitation with temporary credentials.
Alternative Flow	4a. The Super Admin resends the invitation if required.
Exception Flow	3a. If the caller is an Admin without Super Admin role, the system rejects the request.
Priority	High

Use Case Description: UC-A002

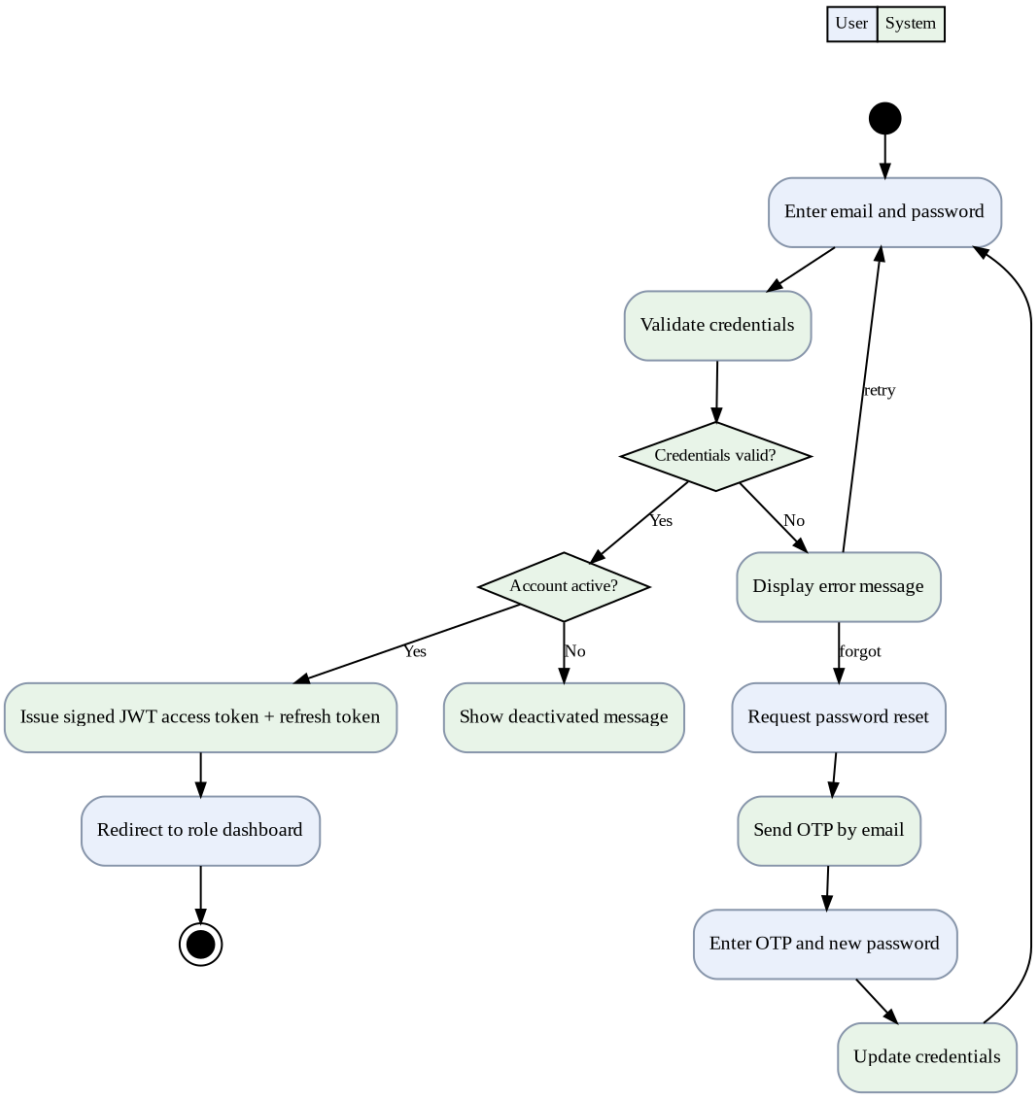
Field	Description
Use Case ID	UC-A002
Use Case Name	Publish Marketplace Product
Primary Actor	Administrator
Description	The administrator publishes an educational material product to the public catalogue.
Preconditions	The administrator is logged in.; A material has been uploaded.
Postconditions	The product has status PUBLISHED and appears in the public catalogue.
Main Flow	1. The administrator creates a product referencing the uploaded material.; 2. The administrator enters title, category, subject, grade, price, and discount.; 3. The system validates pricing rules and saves the product as a draft.; 4. The administrator changes the status to published.; 5. The product appears in the public catalogue.
Alternative Flow	2a. The administrator marks the product as free; the price is set to zero.
Exception Flow	3a. If a paid product has no positive price, the system rejects the request.
Priority	Medium

Chapter 7: Activity Diagrams

Activity diagrams illustrate the dynamic workflows and operational processes of the Mentify Digital Learning Platform. These diagrams represent the sequence of actions, decision points, system validations, and outcomes involved in key user interactions and system operations. Each step is shaded according to the actor or system responsible, as shown in the legend at the top of each diagram.

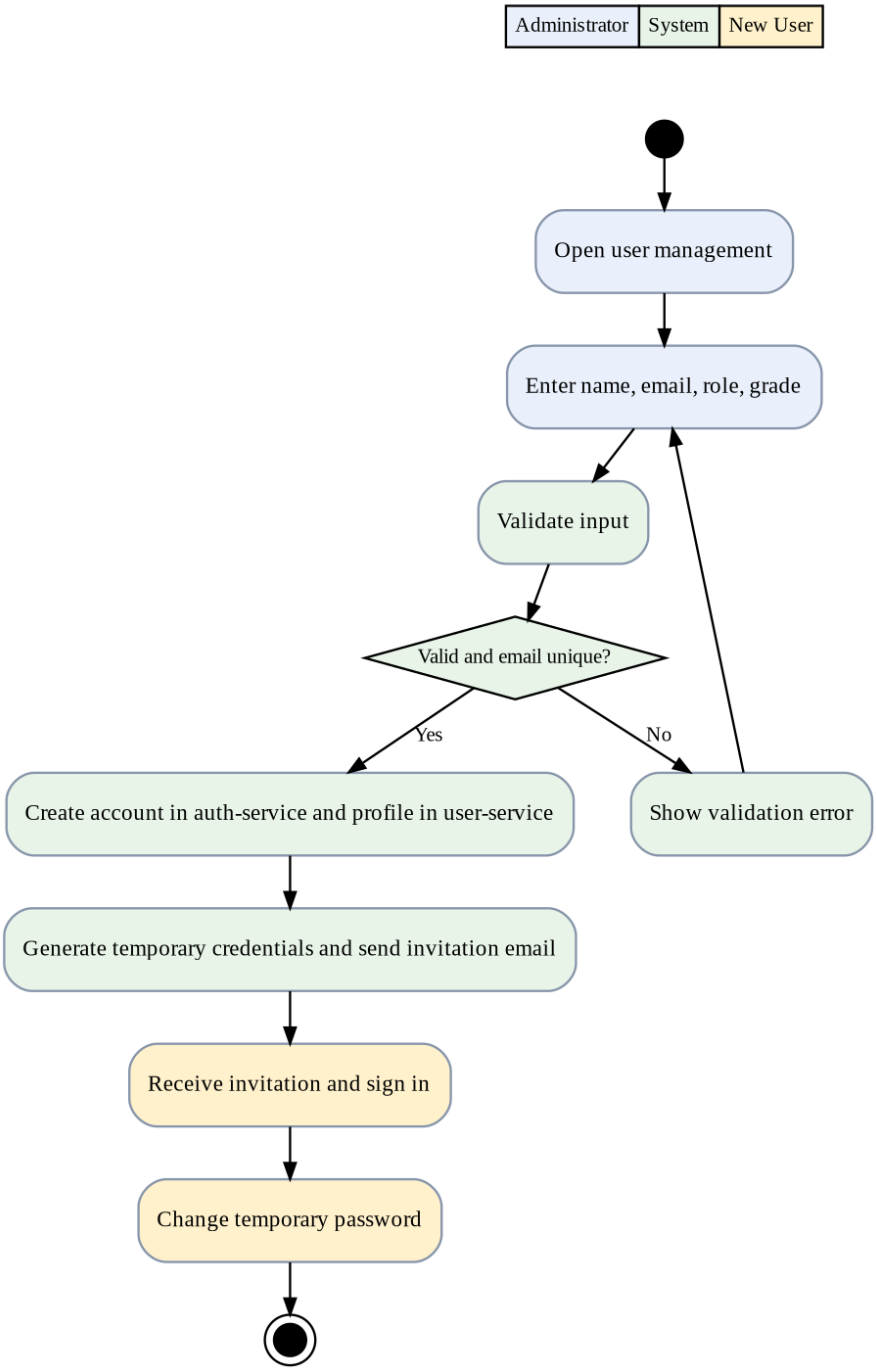
Each activity diagram provides a visual representation of system behaviour for major functional scenarios involving Students, Teachers, Administrators, Super Administrators, and the AI subsystem. The Support Team Handles Ticket and AI Help Chatbot diagrams of the original document have been replaced by the Course Payment and AI Conversation diagrams, which reflect the implemented system.

7.1 Student Login (JWT-based)



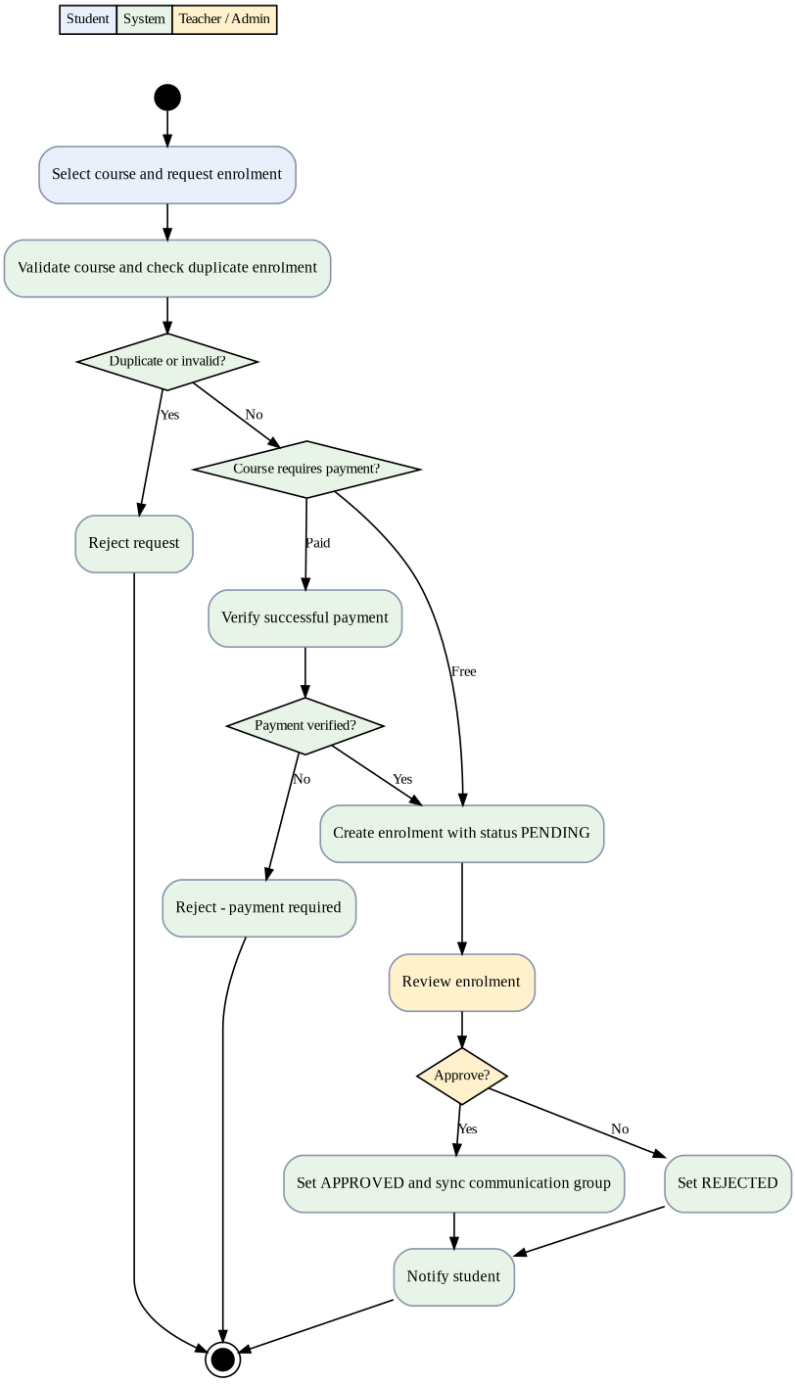
This activity diagram illustrates the login workflow. The user enters an email and password and the system validates the credentials. If they are invalid, an error message is displayed with options to retry or start password recovery, in which an OTP is sent by email and a new password is set. If the credentials are valid, the system checks that the account is active. If the account is deactivated, a message is displayed. Otherwise the system issues a signed JWT access token and a refresh token and redirects the user to the dashboard.

7.2 Student Registration



This activity diagram illustrates the registration process. The administrator enters the student's details and the system validates the input and checks that the email is unique. The account is created in the authentication service and the profile in the user service, and an invitation with temporary credentials is emailed. The student signs in and changes the temporary password. Third-party (Google) registration is not part of the implemented flow.

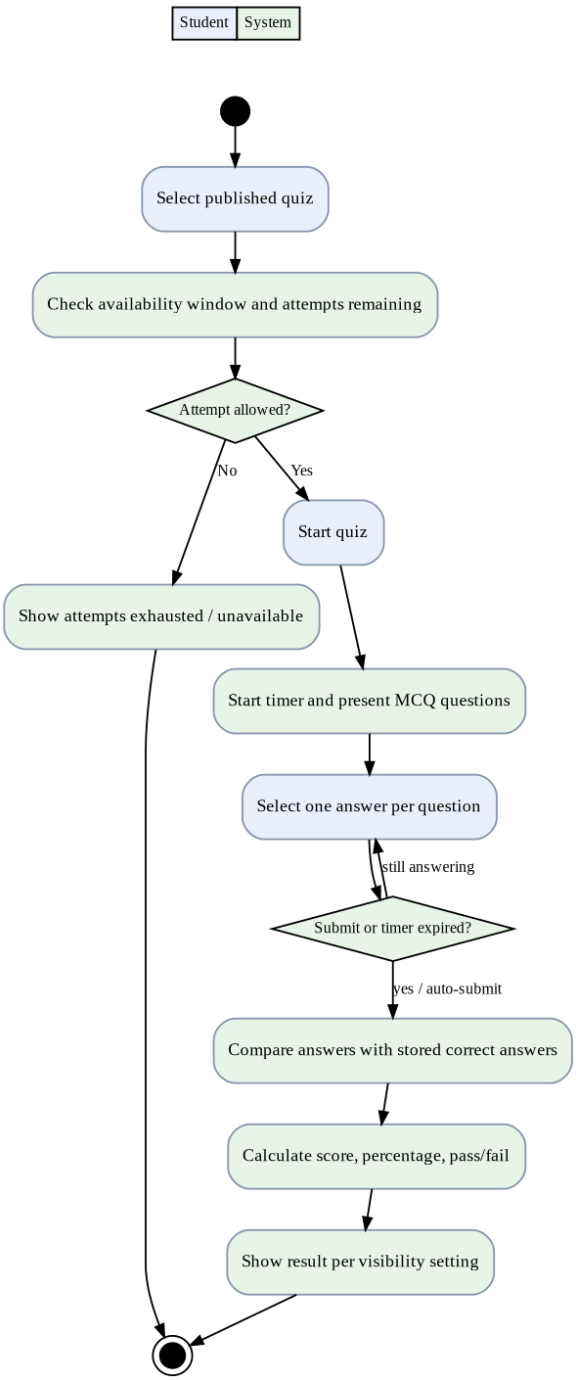
7.3 Enrol in Course



This activity diagram describes the enrolment workflow. The student requests enrolment and the system validates the course and checks for duplicate enrolment. For a paid course, the system verifies that a successful payment exists and rejects the request otherwise. The enrolment is created with status PENDING. An authorised teacher or administrator sets the status to APPROVED or REJECTED, communication-group membership is synchronised

on approval, and the student is notified.

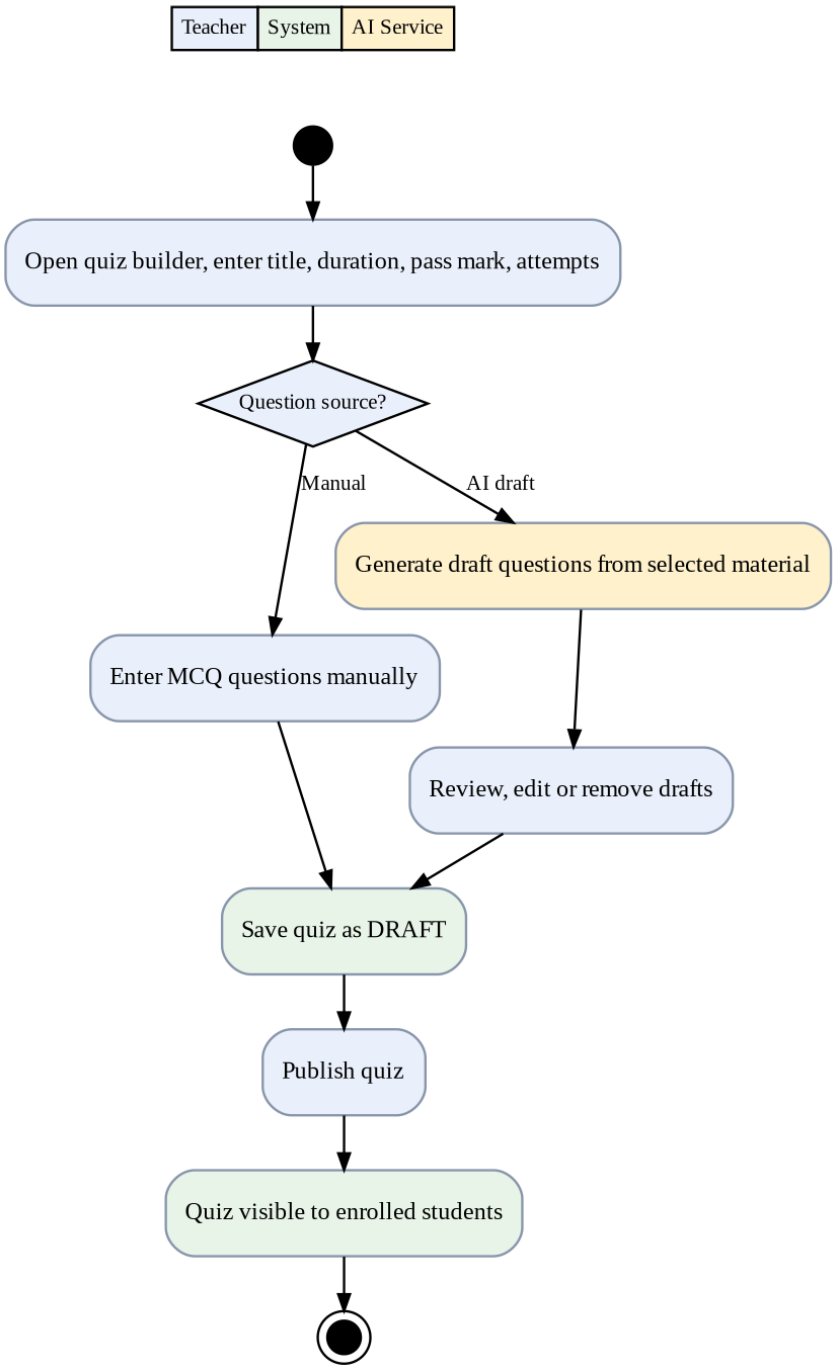
7.4 Attempt Online Quiz



This activity diagram illustrates the quiz-taking process. The system verifies the quiz availability window and the remaining attempts. If an attempt is allowed, the student starts the quiz, the system starts the timer, and the student selects one answer per question. When the

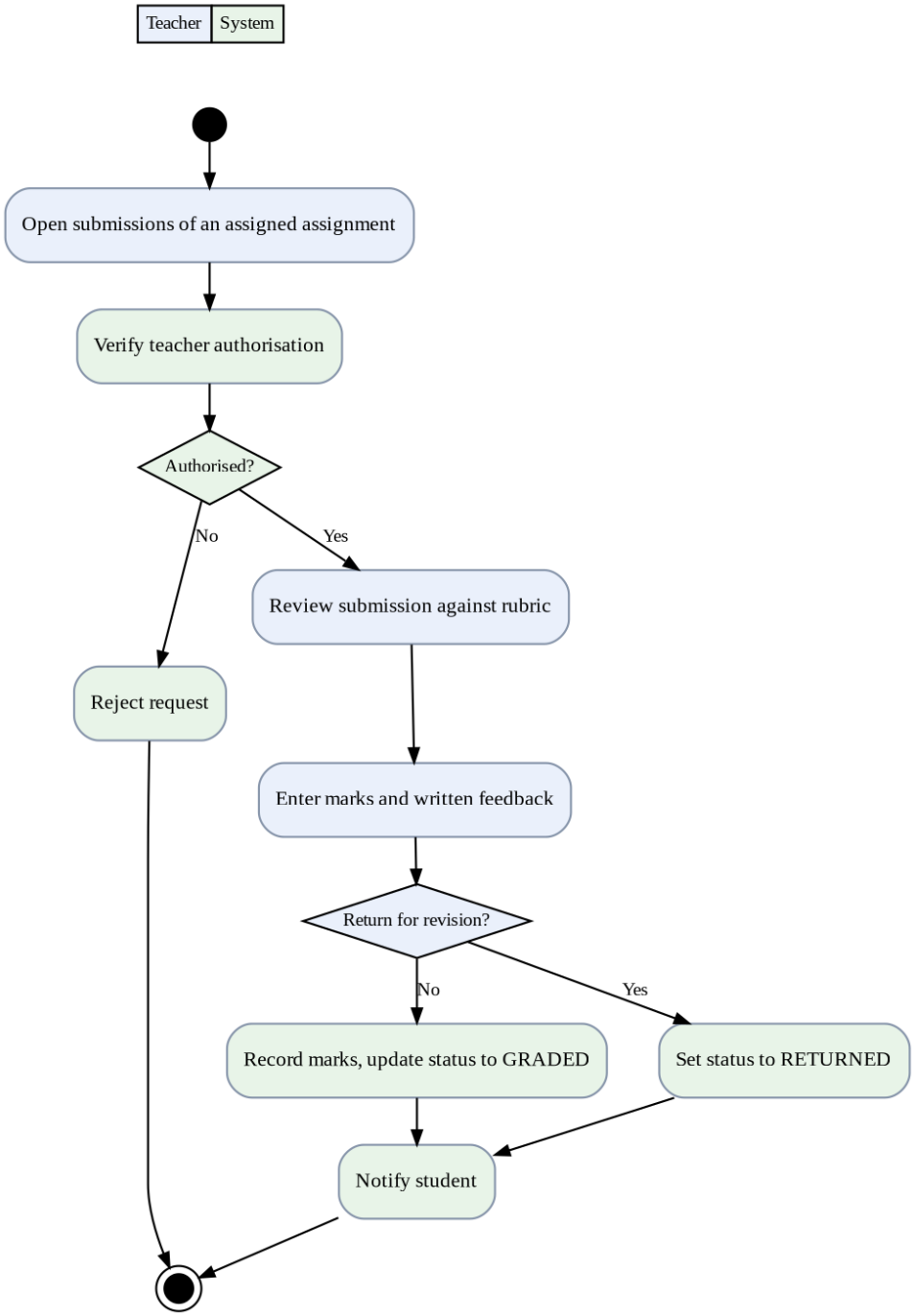
student submits or the timer expires, the system compares the answers with the stored correct answers, calculates the score, percentage, and pass/fail outcome, and shows the result according to the visibility setting.

7.5 Teacher Creates and Publishes Quiz with AI Assistance



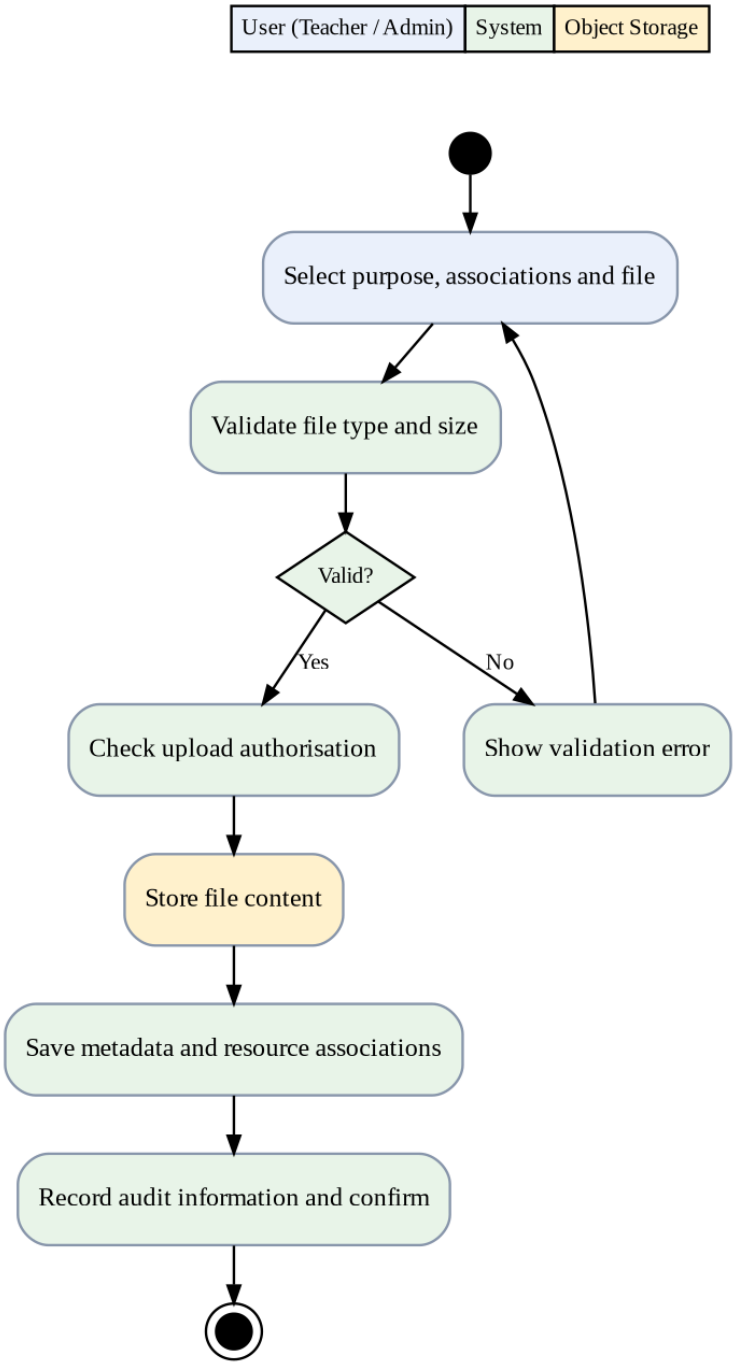
This activity diagram represents the quiz creation workflow. The teacher enters the quiz details and either types MCQ questions manually or requests AI-generated draft questions. The teacher reviews, edits, or removes the drafts. The quiz is saved in draft status and later published to enrolled students.

7.6 Teacher Grades Assignment



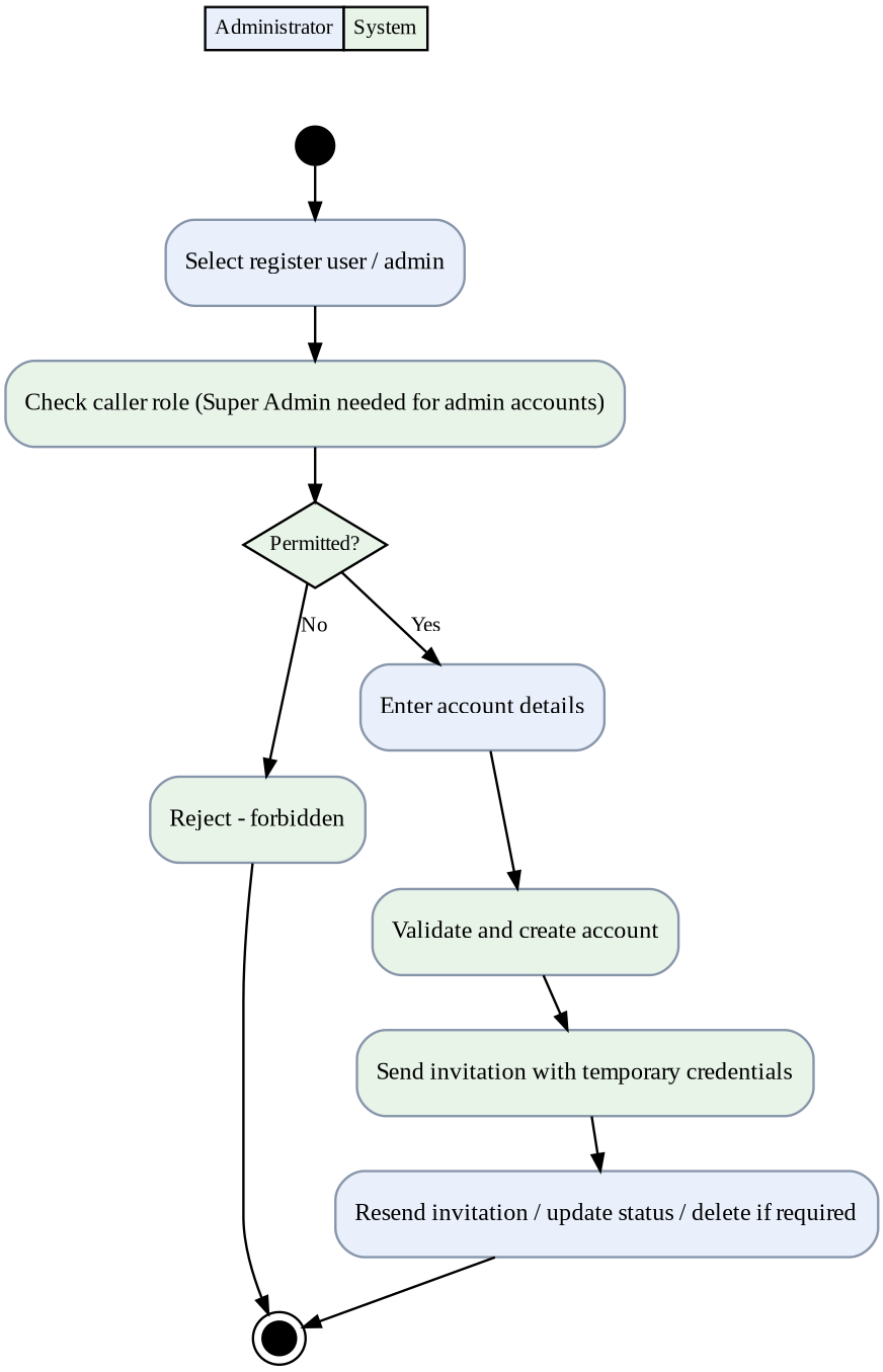
This activity diagram shows the assignment grading process. The teacher opens the submissions of an assigned assignment and the system verifies authorisation. The teacher reviews the submission against the rubric and enters marks and written feedback. The system records the grade or returns the submission for revision and notifies the student. Assignment scoring is performed by the teacher; AI grading is deferred.

7.7 Upload Learning Materials



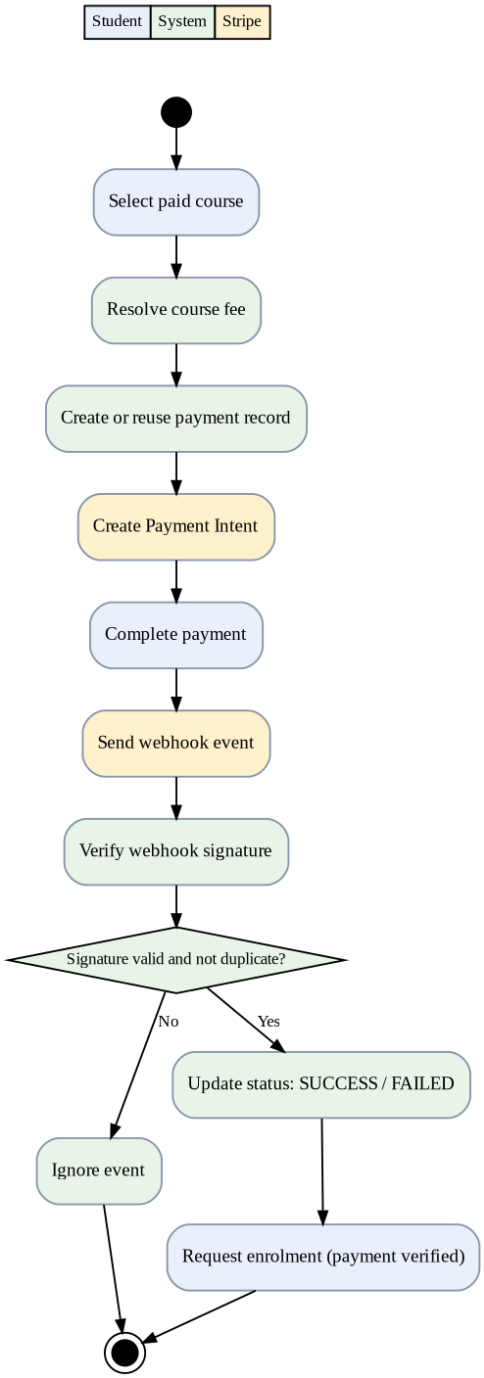
This activity diagram illustrates the content upload workflow. The user selects the file purpose and associations and uploads the file. The system validates file type and size and checks authorisation. The file content is stored in object storage, metadata and resource associations are saved, audit information is recorded, and confirmation is provided.

7.8 Admin Creates User Account



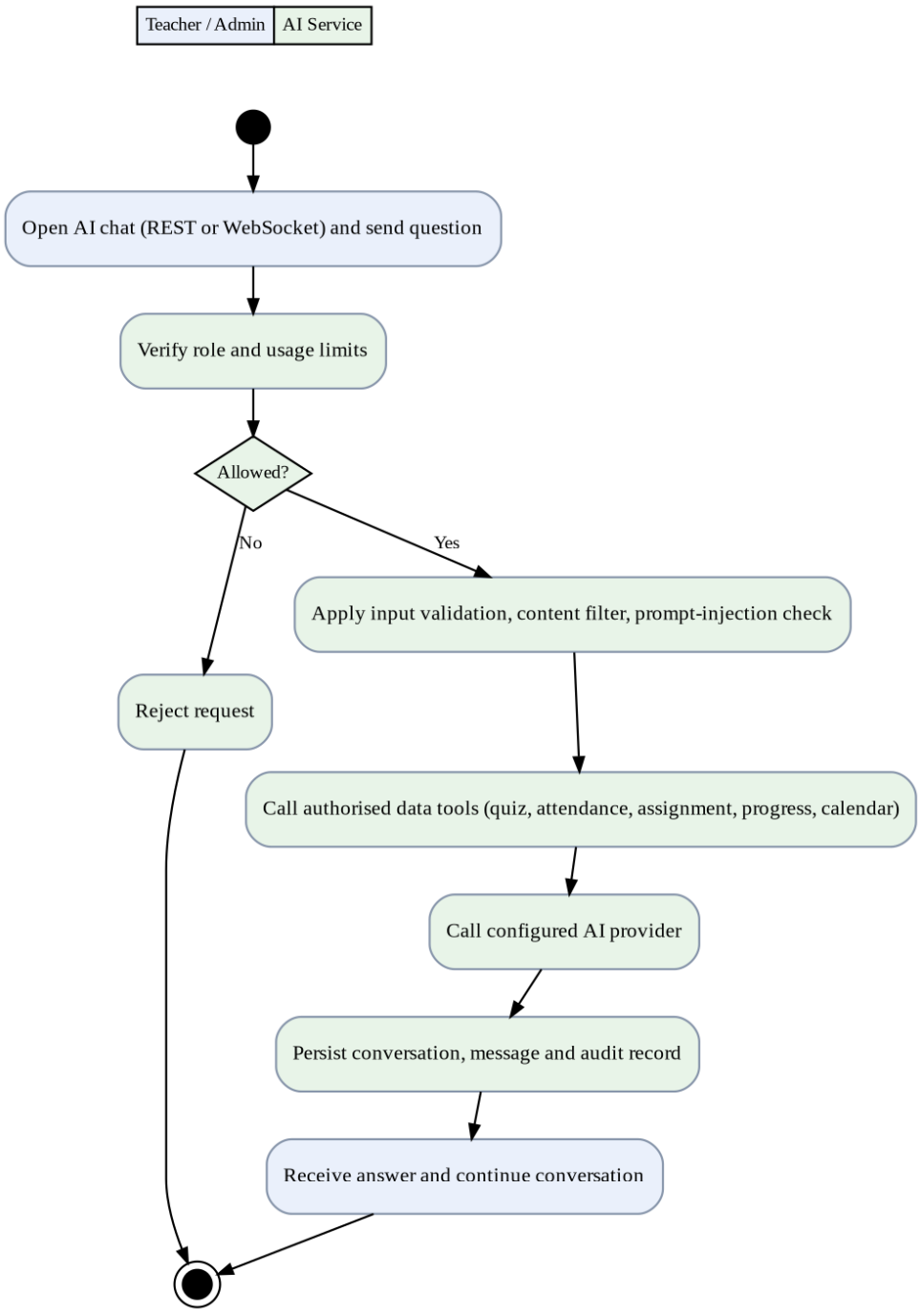
This activity diagram shows the administrative account creation process. The system checks the caller's role; administrator accounts require the Super Admin role. The administrator enters the details, the system validates them, creates the account, and sends an invitation with temporary credentials. The administrator may also resend the invitation, update the account status, or delete the user.

7.9 Course Payment (Stripe)



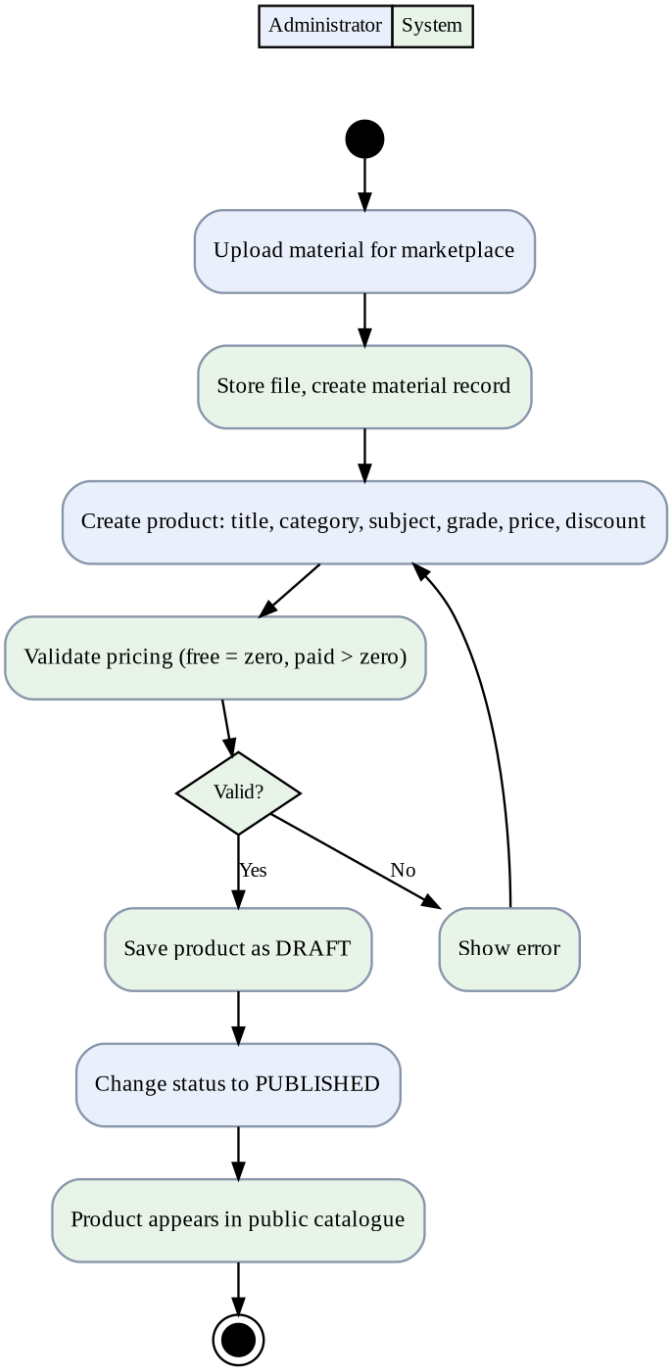
This activity diagram describes the course payment lifecycle. The system resolves the course fee, creates or reuses a payment record, and creates a Stripe Payment Intent. After the student completes payment, Stripe sends a webhook event whose signature is verified. Duplicate events are ignored, and the payment status is updated to success or failed. The student may then request enrolment.

7.10 AI Conversation



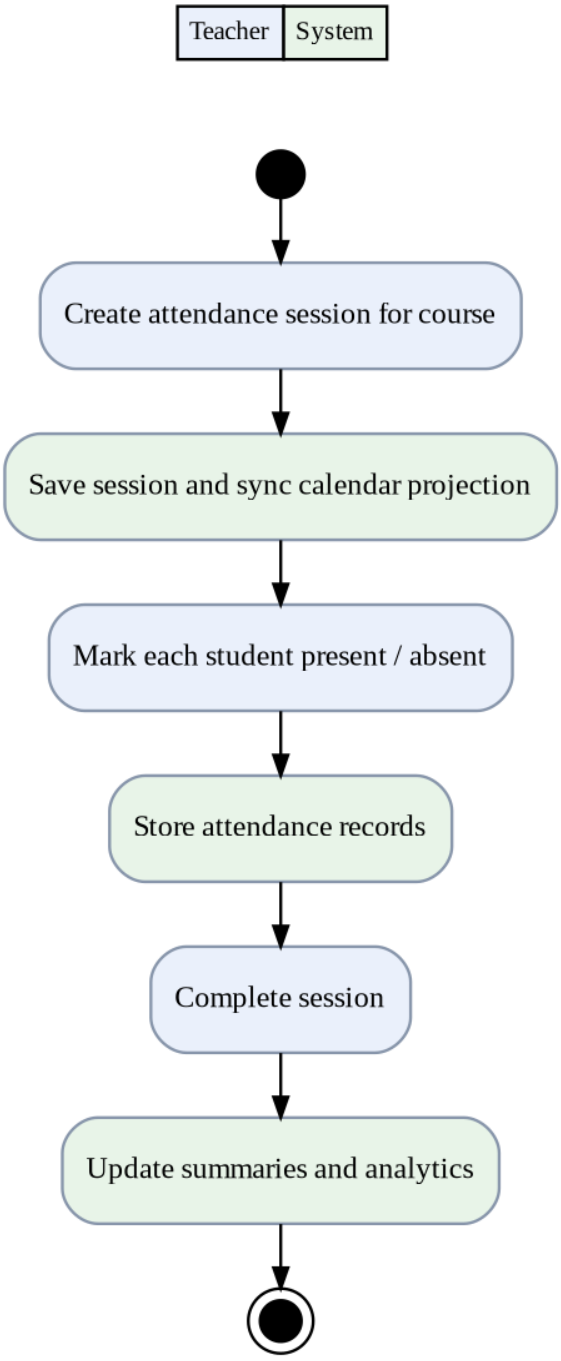
This activity diagram represents the conversational AI interaction for Teacher, Admin, and Super Admin roles. The system verifies the role and usage limits, applies input validation, content filtering, and prompt-injection checks, retrieves permitted data through authorised tools, calls the configured AI provider, and persists the conversation, message, and audit record before returning the answer.

7.11 Marketplace Product Publication



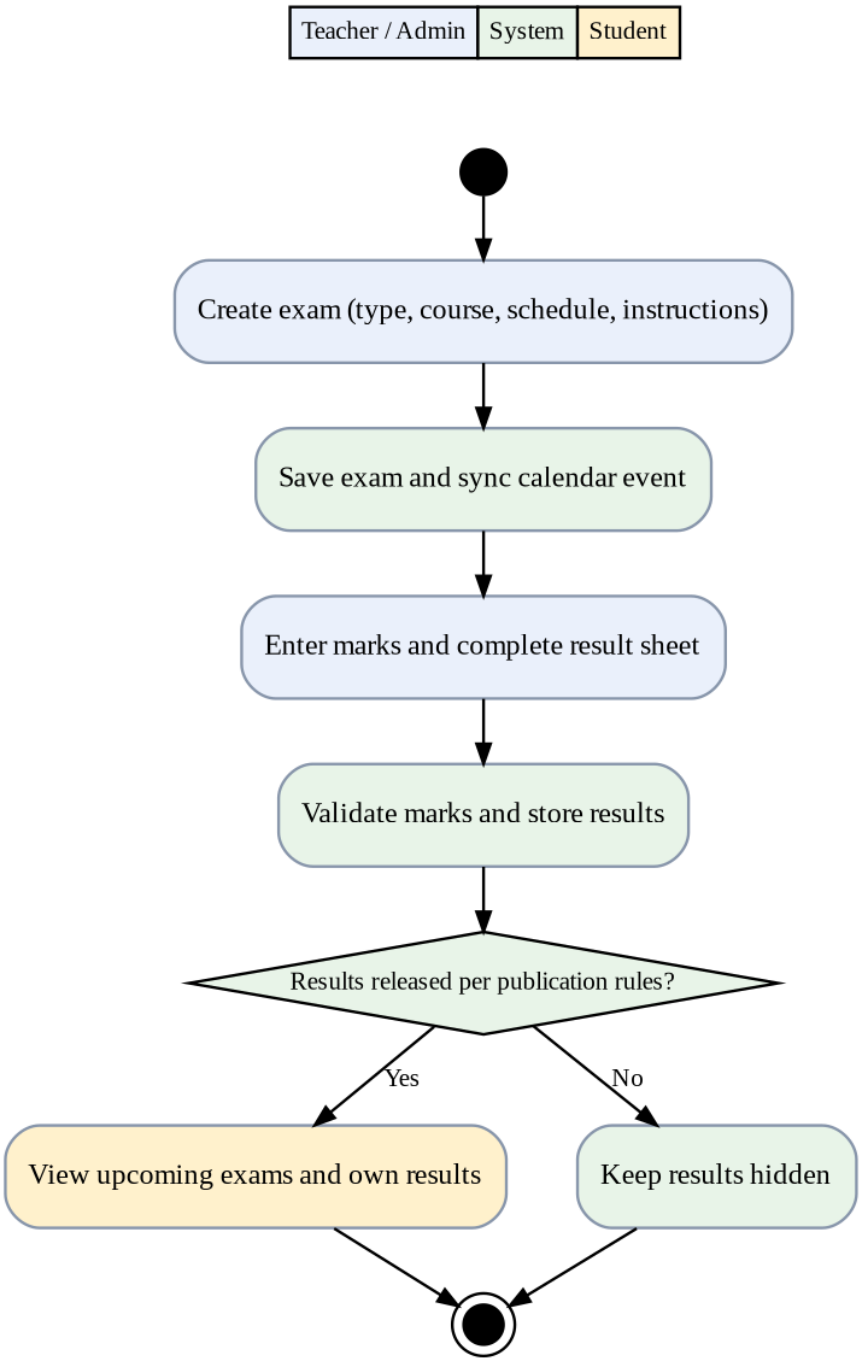
This activity diagram describes the marketplace publication workflow. The administrator uploads a material, creates a product referencing it, and the system validates the pricing rules (free products have a zero price; paid products have a positive price). The product is saved as a draft and appears in the public catalogue after it is published. Checkout is not part of this flow.

7.12 Attendance Recording



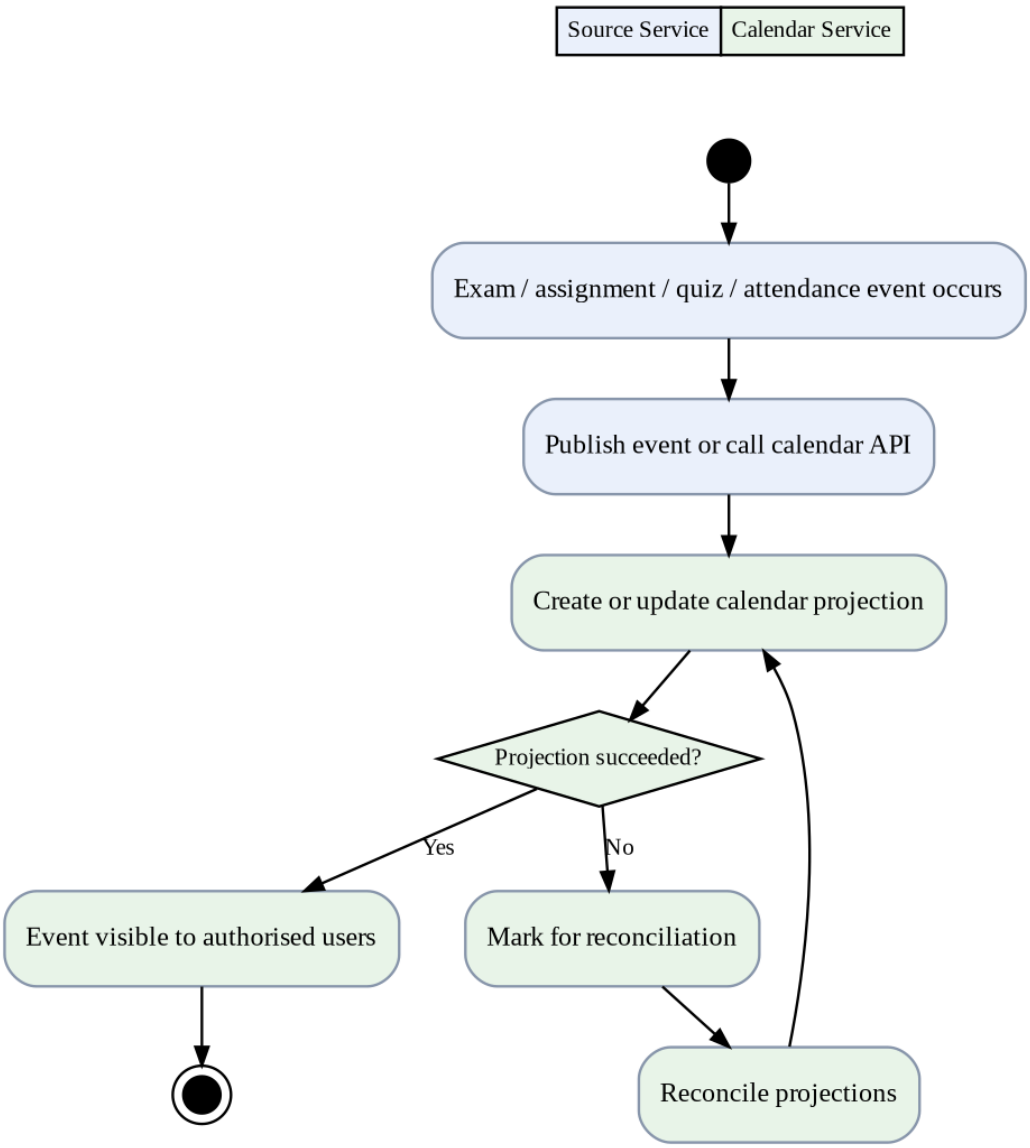
This activity diagram describes attendance recording. The teacher creates an attendance session, the system saves it and synchronises a calendar projection, the teacher marks each student, the system stores the records, and on completion updates attendance summaries and analytics.

7.13 Examination Management



This activity diagram describes examination management. An authorised user creates an examination, which is synchronised with the calendar. Marks are entered and validated, and the results are released to students according to the publication rules.

7.14 Calendar Synchronisation



This activity diagram describes how examination, assignment, quiz, and attendance events are reflected in the academic calendar. The source service publishes the event, the calendar service creates or updates its projection, and a failed projection is marked for reconciliation and retried.

Chapter 8: External Interface Requirements

This chapter describes the external interfaces of the Mentify Digital Learning Platform, including user interface characteristics, hardware interactions, software integrations, and communication protocols required for system operation.

8.1 User Interface Requirements

The system shall provide a modern, responsive, web-based user interface that is intuitive, accessible, and aligned with contemporary usability standards. The interface shall adopt a clean and minimalist design with consistent colour theming that reflects the branding guidelines of TIT Education.

The system interface shall be fully responsive and capable of adapting seamlessly across desktop, tablet, and mobile devices. Accessibility requirements shall comply with WCAG 2.1 Level AA standards, including appropriate colour contrast ratios, keyboard navigation support, screen reader compatibility, and alternative text for all non-text content.

The dashboard shall utilise visual elements such as progress indicators, charts, and card-based layouts to present information in a clear and structured manner. All input forms shall implement client-side validation with descriptive and user-friendly error messages. Navigation shall be role-based, ensuring that users only view features and menu options relevant to their assigned roles (Student, Teacher, Admin, or Super Admin), while the back-end enforces authorisation independently of the interface.

8.2 Hardware Interface Requirements

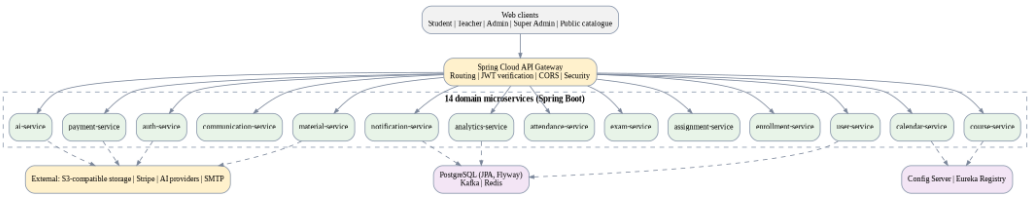
The Mentify Digital Learning Platform is designed as a cloud-hosted web application and therefore does not require direct integration with specialised hardware components. However, the system shall support interaction through standard input devices, including keyboards, mice, and touch-enabled interfaces.

The platform shall be compatible with display devices ranging from small mobile screens (approximately 320 pixels in width) to high-resolution displays up to 4K resolution. The system shall support file uploads from locally connected storage devices available on

the user's machine.

8.3 Software Interface Requirements

The system shall integrate with external software services and application programming interfaces to support authentication, payments, AI-powered functionality, storage, messaging, and data management. The overall structure of the backend is shown in the architecture overview below, which is a conceptual grouping rather than a literal service dependency diagram.



Authentication shall be provided by the first-party authentication service, which issues and validates signed JWTs. Third-party Google OAuth 2.0 and Keycloak are not part of the implemented system. Course payments shall be processed through the Stripe API using Payment Intents, webhooks, and refunds. AI-driven functionalities, including content summarisation, quiz draft generation, and conversational assistance, shall be implemented through configurable AI service providers such as Gemini or OpenAI, subject to implemented adapters and runtime configuration.

The system shall interface with email delivery through SMTP-based providers to facilitate transactional communications, including invitations, password reset OTPs, and notifications. S3-compatible object storage shall be used for storing learning materials, assignment submissions, and related content. The platform shall interface with PostgreSQL for structured data storage across its services, including the AI service; MongoDB is not used as the AI database. The interfaces are summarised below.

Interface	Purpose
Frontend to API Gateway	Client access to backend APIs over HTTPS
Gateway to domain services	Request routing and JWT security integration
Service-to-service REST (OpenFeign)	Synchronous domain integration

Interface	Purpose
Kafka messaging	Asynchronous events between services
WebSocket / STOMP	Real-time communication and AI chat
PostgreSQL	Persistent application data
S3-compatible object storage	File content storage
Stripe API	Course payment operations and webhooks
AI provider APIs	AI-assisted functionality
SMTP / email	Invitations, OTP delivery, and notifications
Eureka	Service registration and discovery
Config Server	Centralised service configuration

Layer	Implementation
Backend	Java 17, Spring Boot 3.2.5
API security	Spring Security, JWT, OAuth2 Resource Server
Persistence	PostgreSQL, JPA/Hibernate, Flyway migrations
Service discovery and configuration	Netflix Eureka, Spring Cloud Config, Spring Cloud Gateway
Inter-service calls	OpenFeign
Real-time communication	WebSocket/STOMP
Event messaging	Kafka
Caching and coordination	Redis
File storage	S3-compatible object storage
Payments	Stripe
Containerisation	Docker, Docker Compose, Caddy reverse proxy

8.4 Communication Interface Requirements

All communication between client and server components shall be conducted over secure HTTPS connections using TLS version 1.2 or higher. The system shall expose RESTful application programming interfaces that use JSON as the primary data exchange format.

Real-time communication features, including group messaging and AI chat, shall be implemented using the WebSocket protocol with STOMP to ensure low-latency data transmission. Asynchronous inter-service events shall be exchanged through Kafka. The system APIs shall conform to REST architectural principles and utilise appropriate HTTP status codes for all responses, including conflict responses where late submission or attempt

limits are violated.

File transfer operations shall support multipart/form-data encoding for efficient handling of uploaded content, and file downloads shall use time-limited authorised URLs. The system shall also implement rate-limiting and usage-limiting mechanisms, including AI usage limits, to prevent misuse, ensure service stability, and enhance overall security.

Chapter 9: Data Requirements

This chapter describes the types of data that shall be managed, processed, stored, and protected by the Mentify Digital Learning Platform. The system shall handle multiple categories of data essential for learning delivery, assessment, payments, communication, and analytics. Application data is persisted in PostgreSQL by the persistence-enabled domain services, while uploaded file content is stored separately in S3-compatible object storage; the material service holds the metadata and authorisation rules.

9.1 User Data

The system shall store and manage user-related data, including personal identification information such as name, email address, grade level, and assigned role (Student, Teacher, Admin, or Super Admin). Authentication data, including securely hashed passwords, refresh sessions, and issued JWT identities, shall be owned by the authentication service, while profile information such as profile photographs, contact details, and parental contact information where applicable shall be owned by the user service. The system shall also maintain account status and invitation information.

9.2 Course Data

The system shall maintain academic grade data, including grade definitions and grade-level monthly fees, and course-related data, including course name, course code, description, subject, grade, assigned teacher, academic year, semester, level, capacity, credits, schedule, grading weights, teaching assistants, and online, visible, and published flags. It shall store structured content organisation, including modules and lessons, along with associated learning materials such as documents, multimedia files, and videos. Enrolment data shall include the student, course, status (PENDING, APPROVED, or REJECTED), and timestamps.

9.3 Assessment Data

The platform shall manage assessment-related data, including assignment specifications such as instructions, start and due dates, maximum marks, allowed attempts, late-submission rules, penalty percentage, and rubric items. It shall store quiz configurations, including single-answer MCQ question sets, answer options, correct answers, duration, pass mark, maximum attempts, and start and end times. Student submission data shall include uploaded files, textual responses, attempt counts, and submission timestamps. The system shall also record marks, scores, quiz results, and written feedback provided by teachers. Examination data shall include examination type, schedule, instructions, attachments, marks, and result sheets, and attendance data shall include sessions, status, and student records.

9.4 Communication Data

The system shall store communication-related data, including communication groups, group membership, group messages, message attachments, communication settings, announcements, and system-generated notifications with their delivery information. AI conversation data, including conversations, messages, and request audit records, shall also be persisted. This data shall support collaborative learning and information dissemination across the platform.

9.5 Analytics Data

The system shall collect and maintain analytics-related data derived from domain events, including enrolment statistics, assignment and quiz performance, examination results, attendance records and summaries, registration statistics, and payment analytics. Performance analytics shall include score trends and learning progress indicators. Calendar data, including events, venues, and projections from examinations, assignments, quizzes, and attendance sessions, shall be maintained by the calendar service.

9.6 Payment and Marketplace Data

Payment-related data shall include payment records, student and course references, amounts and currency, the Stripe payment identifiers, payment status (pending, success, failed, or refunded), refund records, and processed webhook event identifiers used to prevent duplicate processing. Marketplace data shall include product records with material reference, title, descriptions, category, subject, grade, price, free flag, discount percentage, currency, publication status, thumbnail, and timestamps. Card details are processed by Stripe and shall not be stored by the system.

Data Domain	Owning Service	Storage
Credentials, refresh sessions	auth-service	PostgreSQL
User profiles	user-service	PostgreSQL
Grades, fees, courses, modules, lessons	course-service	PostgreSQL
Enrolments	enrollment-service	PostgreSQL
Assignments, submissions, quizzes, grades	assignment-service	PostgreSQL
Examinations and results	exam-service	PostgreSQL
Attendance	attendance-service	PostgreSQL
Material metadata, marketplace products	material-service	PostgreSQL + object storage (file content)
Payments and refunds	payment-service	PostgreSQL
AI conversations and audit records	ai-service	PostgreSQL
Groups and messages	communication-service	PostgreSQL
Notifications	notification-service	PostgreSQL
Calendar events and projections	calendar-service	PostgreSQL
Analytics	analytics-service	PostgreSQL

Logical database separation per service shall be maintained; this does not require physically separate database servers.

9.7 Data Security and Retention

All personal and sensitive data shall be encrypted at rest using AES-256 encryption standards where supported by the hosting environment (target; verification pending). Data in transit shall be protected using TLS 1.2 or higher. Data retention policies shall comply with the institutional policies of TIT Education as well as applicable data protection regulations. Automated database backups shall be performed at intervals of 24 hours and retained for a minimum period of 30 days to ensure data recoverability and system reliability.

Chapter 10: Alternative Solutions Considered

During the system planning and requirement analysis phase, several alternative implementation approaches were evaluated to determine the most suitable solution for developing the Mentify Digital Learning Platform.

10.1 Adoption of an Existing Learning Management System

One potential approach considered was the adoption of an existing Learning Management System such as Moodle, Canvas, or Google Classroom. These platforms provide pre-built functionalities for course management, assessment delivery, and communication features. However, they lack the advanced AI-driven capabilities and the integrated grade-based fee, payment, and enrolment workflow required by TIT Education. Furthermore, these platforms offer limited flexibility for customisation to align with the institution's specific workflows and pedagogical strategies. Proprietary platforms also involve recurring licensing costs. Although Moodle is open-source, it would require extensive custom development to support the desired AI-based functionality and tailored system features.

10.2 Low-Code / No-Code Development Platforms

Another alternative involved the use of low-code or no-code development platforms such as Bubble or AppSheet. This approach could potentially accelerate the development timeline and reduce initial implementation complexity. However, such platforms impose constraints on system performance, scalability, and the integration of advanced AI and payment capabilities. Additionally, reliance on proprietary development environments may introduce long-term challenges in maintainability, system portability, and vendor dependency.

10.3 Custom System Development (Selected Approach)

The selected approach for the Mentify Digital Learning Platform is full custom system development using a microservice architecture. This approach enables complete control over system functionality, user experience design, AI integration, payment integration, and

architectural scalability. Although custom development requires greater initial development effort, it provides the flexibility needed to address TIT Education's requirement for a student-centred, AI-enhanced digital learning environment. Moreover, a custom-built solution allows future expansion and scalability beyond the institutional context. Significant architectural decisions taken during development are to be recorded as ADRs in the SDS.

Chapter 11: Feasibility Study

This chapter evaluates the feasibility of developing and implementing the Mentify Digital Learning Platform across technical, economic, operational, and schedule dimensions.

11.1 Technical Feasibility

The system is technically feasible, and the implemented backend demonstrates this. The selected technology stack, comprising React.js for the frontend, Java and Spring Boot microservices for the backend, PostgreSQL for database management, Kafka and Redis for messaging and caching, S3-compatible object storage, and cloud hosting through AWS or an equivalent provider, is mature, well-documented, and widely supported within the software development ecosystem.

AI-driven functionalities are implemented through integration with established AI service providers, and payments through Stripe. Core architectural components including JWT-based authentication, Role-Based Access Control (RBAC), and WebSocket-based real-time communication are supported by robust frameworks and libraries. Furthermore, the development team possesses the required competencies in web application development, database management, and API integration, ensuring that the technical requirements of the system can be successfully addressed.

11.2 Economic Feasibility

The project demonstrates strong economic feasibility as it is being developed within an academic context, thereby significantly reducing labour-related costs. Infrastructure expenses can be managed effectively through the utilisation of cloud provider free tiers and educational support programmes such as AWS Educate and Google Cloud for Education.

Primary anticipated costs include cloud hosting services, AI API usage, payment processing fees charged by Stripe, and domain registration. These expenses are expected to remain within a manageable budget. From the perspective of TIT Education, the implementation of this system eliminates the need for costly commercial Learning Management System subscriptions and reduces administrative workload through automation, resulting in

long-term cost savings.

11.3 Operational Feasibility

Operational feasibility is supported by the positive reception of the system concept among the staff of TIT Education, who have indicated readiness to adopt the proposed platform. The system is designed with a strong emphasis on usability, requiring only basic digital literacy from its users.

The availability of training materials will facilitate user onboarding and system adoption. Administrators perform user support in the current release, since no dedicated support portal exists. The system is intended to operate alongside existing manual processes during the transition phase, enabling gradual adoption without significant operational disruption.

11.4 Schedule Feasibility

The development of the Mentify Digital Learning Platform aligns with the academic schedule defined by the SENG 31242 module and the SENG 34213 System Development Project. The project is structured into phased development stages, prioritising essential system components such as authentication, course management, content delivery, enrolment, payment, and assessment functionalities.

Features identified as deferred in this document, such as a support portal, AI-based recommendations, and marketplace checkout, are planned for subsequent iterations. The development team, consisting of four members, has distributed responsibilities to support parallel development activities. The adoption of Agile development practices, including two-week sprint cycles, facilitates continuous progress monitoring and supports timely project completion.

Chapter 12: Risk Analysis

This chapter identifies potential risks associated with the development, deployment, and operation of the Mentify Digital Learning Platform. Each risk is evaluated based on its likelihood and impact, and appropriate mitigation strategies are defined to minimise potential adverse effects.

Risk ID	Description	Likelihood	Impact	Risk Level	Mitigation Strategy
R-001	AI API service outages or rate limiting may disrupt AI features such as summarisation, quiz generation, and AI conversation.	Medium	High	High	Keep AI isolated in its own service with timeouts and retries so the platform remains functional without AI features; apply usage limits; evaluate multiple providers for fallback.
R-002	Scope creep due to evolving client requirements may delay the project timeline.	High	High	High	Establish a formal change request process, prioritise features using MoSCoW, and record changes in the Requirement Change Register.
R-003	Team member unavailability due to academic or personal commitments.	Medium	Medium	Medium	Cross-train team members across modules, maintain documentation, and use version control with frequent commits.

Risk ID	Description	Likelihood	Impact	Risk Level	Mitigation Strategy
R-004	Data breach or security vulnerability may expose sensitive student information.	Low	High	High	Conduct regular security audits, encrypt data at rest and in transit, follow OWASP guidelines, enforce role-based and ownership-based authorisation, and protect payment and file access.
R-005	Poor user adoption by teachers and students unfamiliar with digital learning platforms.	Medium	Medium	Medium	Design an intuitive user interface, provide onboarding tutorials, and conduct structured user training sessions.
R-006	Performance degradation under high concurrent user load during peak usage periods.	Medium	High	High	Use Redis caching, scale services independently, conduct stress testing before deployment, and optimise database queries using indexing.
R-007	Failure of third-party services such as Stripe, email services, AI providers, or object storage.	Low	Medium	Medium	Design retry mechanisms, verify webhook events, apply timeouts, and degrade features gracefully.

Risk ID	Description	Likelihood	Impact	Risk Level	Mitigation Strategy
R-008	Insufficient testing coverage leading to undetected defects in production.	Medium	Medium	Medium	Maintain coverage targets above 80%, implement automated unit, integration, and end-to-end testing, and conduct user acceptance testing with TIT Education prior to deployment.
R-009	Database corruption or data loss resulting from system failure.	Low	High	Medium	Implement automated daily backups with geographic redundancy, enable point-in-time recovery, and regularly test restoration procedures.
R-010	Regulatory compliance issues related to student data handling.	Low	High	Medium	Align with institutional data policies, anonymise data for analytics, obtain parental consent where required, and document all data processing activities.

Risk ID	Description	Likelihood	Impact	Risk Level	Mitigation Strategy
R-011	Payment errors such as duplicate webhook events, failed payments, or incorrect fee amounts.	Medium	High	High	Verify webhook signatures, process events idempotently, restrict payment access by ownership, and test payment and refund flows.
R-012	Operational complexity of a microservice architecture (service coordination, configuration, and deployment).	Medium	Medium	Medium	Use centralised configuration, service discovery, health checks, Flyway migrations, and containerised deployment.
R-013	The SRS may become inconsistent with the implementation as development continues.	Medium	Medium	Medium	Maintain the traceability matrix, record changes in the change register, and review the SRS with the supervisor at each milestone.
R-014	Requirements marked Not Verified or Partially Implemented may be mistaken for completed features.	Medium	Medium	Medium	Use the Status column consistently and update statuses only with implementation and test evidence.

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Appendix A: Requirement Change Register

This register records every significant change made to the original (version 01) requirements so that original commitments remain traceable. Superseded or deferred items are not deleted from the specification history.

Original Requirement / Claim	Action	Reason	Updated Reference
Payment processing and financial management excluded (Section 1.2)	Updated	The implemented system includes Stripe-based course payments, payment history, and refunds.	Section 1.2; FR-PY001–PY009
Five user categories incl. Support Team and AI System (Section 2.1)	Updated	Implemented roles are Student, Teacher, Admin, and Super Admin; AI is a supporting subsystem.	Sections 2.1, 2.2; BR-006
FR-S001 / UC-S001 Google authentication; Section 8.3 Google OAuth	Removed / Deferred	Authentication is first-party JWT; no Google sign-in is established.	FR-S001; Section 8.3
FR-S004 change password / FR-T004 forgot password	Updated	Custom OTP-based recovery flow.	FR-S004; FR-AU005

Original Requirement / Claim	Action	Reason	Updated Reference
FR-S008 enrol in course; UC-S008; Section 7.3 auto-approval	Updated	Student enrolment is PENDING and paid courses require verified payment.	FR-S008; FR-EN001–EN008; BR-010, BR-011
FR-S010 student drops a course	Deferred	No student-authorised un-enrolment path is established.	FR-S010
FR-S011 / FR-AI001 AI course recommendations	Deferred	Not established in the inspected APIs.	FR-S011; FR-AI001
FR-S017 / FR-S029 / FR-AI003 student AI assistant and 24/7 help chatbot	Updated / Deferred	Conversational AI is restricted to Teacher, Admin, Super Admin; student access is limited to summarisation; ticket escalation is not implemented.	FR-S017; FR-S029; FR-AI002–AI003
FR-S022 MCQ and short-answer quizzes	Updated	Only single-answer MCQ is implemented.	FR-S022; FR-T024; BR-017
FR-S025 AI-generated feedback; FR-T027 / FR-AI005 / FR-AI007 AI grading	Deferred	Assignment grading and feedback are teacher-performed; MCQ scoring is deterministic.	FR-S025; FR-T027; FR-AI005; FR-AI007

Original Requirement / Claim	Action	Reason	Updated Reference
FR-T006 / FR-T007 course creation and deletion by teachers	Updated	Additional course attributes exist; deletion restricted to Admin and Super Admin.	FR-T006; FR-T007; FR-GC003–GC008; BR-014
FR-A002 / FR-A007 assign and create custom roles	Updated / Deferred	Fixed four-role model; custom roles unsupported.	FR-A002; FR-A007; BR-006
FR-A004 SSO; FR-A005 MFA	Deferred / Not Verified	Not established in the implementation.	FR-A004; FR-A005; NFR-SE005
Section 3.4 Support Team requirements FR-SP001–SP011; UC 6.1.4; Activity 7.9	Deferred	No Support role, ticketing portal, or knowledge base in the implementation.	Section 3.4
Section 8.3 MongoDB for chat logs and activity data; push notification service	Updated	AI and chat data use PostgreSQL; push notification service not established.	Sections 8.3, 9.4
NFR-SE002 bcrypt cost factor 10	Updated	Aligned with the development guideline minimum of 12.	NFR-SE002

Original Requirement / Claim	Action	Reason	Updated Reference
Absent: examinations, attendance, calendar, material management, marketplace, analytics, grades and fees	Added	Implemented modules missing from the original SRS.	Sections 3.7–3.16
Absent: refund, webhook, idempotency, AI guardrails, file authorisation requirements	Added	Security-relevant behaviour present in implementation.	NFR-SE006–SE008; FR-PY004; FR-AI010
All functional requirements marked Proposed	Updated	Replaced by implementation Status values.	Chapter 3

Appendix B: Requirements Traceability Matrix

Each requirement group is mapped to implementation evidence in the Mentify backend repository and to the verification still to be recorded. A controller or service existing in the repository is recorded as Implemented in code – testing pending until its behaviour has been validated by test results. This matrix shall be extended to every individual requirement identifier.

Requirement Group	Implementation Evidence	Verification
Roles and RBAC (FR-A002, FR-A006, BR-006)	Role.java (STUDENT, TEACHER, ADMIN, SUPER_ADMIN)	Authorisation tests – pending
JWT authentication (FR-AU001–AU008)	auth-service AuthController and security configuration	Authentication tests – pending
User provisioning (FR-A001, FR-A003, FR-SA001)	user-service registration, invitation resend, status update, and delete endpoints	User API tests – pending
Course and grade management (FR-GC001–GC008)	CourseController, CourseRequest	Course API tests – pending
Student self-enrolment (FR-EN001–EN008)	EntrollmentServiceImpl; enrolment status enumeration	Enrolment integration tests – pending
Stripe payments (FR-PY001–PY009)	PaymentController, PaymentServiceImpl	Stripe integration tests – pending
Assignment grading (FR-T021–T029)	GradingServiceImpl; assignment services	Grading tests – pending
MCQ quizzes and grading (FR-T024–T026, BR-017)	QuestionType (single-answer MCQ), QuizGradingServiceImpl, CreateQuizRequest	Quiz grading tests – pending
AI conversations (FR-AI003, FR-AI008–AI012)	AiConversation, AiChatServiceImpl, chat and WebSocket controllers	AI conversation tests – pending
Material management (FR-MT001–MT009)	MaterialController and retrieval services	Material authorisation tests – pending

Requirement Group	Implementation Evidence	Verification
Marketplace catalogue (FR-MK001–MK006)	MarketplaceProductController, MarketplaceProduct, MarketplaceProductServiceImpl	Catalogue tests – pending
Examinations (FR-EX001–EX006)	exam-service controllers and services	Examination tests – pending
Attendance (FR-AT001–AT005)	attendance-service controllers and services	Attendance tests – pending
Calendar (FR-CA001–CA004)	calender-service event, venue, and projection services	Calendar integration tests – pending
Communication and notifications (FR-CN001–CN005)	communication-service and notification-service	Messaging tests – pending
Analytics (FR-AN001–AN006)	analytics-service and domain analytics controllers	Analytics tests – pending
Microservice architecture (NFR-M001)	services/pom.xml (14 business services), API Gateway, Config Server, Eureka	Architecture review – pending

Client Requirement Sign-Off Page

By signing below, the client representative confirms that the requirements specified in this document accurately reflect the needs of TIT Education, including the deferred items listed in Appendix A.

Name	Designation	Signature	Date
	Client Representative, TIT Education Project Supervisor Project Team Representative		